



ECON 100B: Macroeconomics

Class 21: Stabilization policy [2/2]
(Jones Chapter 13)

Ryan D. Edwards
April 14, 2026

Agenda

- [April 2026 IMF World Economic Outlook](#)
- Go through the class20pollev slides to clarify what I screwed up in the Poll Everywhere questions last time. Slides are on the calendar
- Finish up Chapter 13: Stabilization Policy
 - Central banks often state **inflation targets** like $\bar{\pi} = 0.02$
 - A **monetary policy rule** is a formal statement about how the central bank will raise or lower interest rates in reaction to π and $\tilde{Y}$
 - Both of these (and other techniques) aim to **establish credibility**

Same announcements

- **Problem Set 3** is available, due on **Friday, April 17**
- **Problem Set 4** will be due on **Friday, May 1**
- The **Final exam** is confirmed for **Monday, May 11**
 - Just like at midterm, the main class meets here in **Pimentel 1**
 - We will run 90 minutes from about 3:00-4:30PM
 - DSP students: Please either make a proctoring appointment, or for 150% time only, we will have **Anthropology and Art Practice 160**
(I think it's kind of near Caffe Strada on Bancroft)

What is the IMF?

- The International Monetary Fund (IMF) and the World Bank (IBRD) were devised in [July 1944 at Bretton Woods](#)
- The broad ideas
 - To avoid mistakes after World War 1
 - To leverage new macroeconomic insights from John Maynard Keynes
- The IMF usually provides emergency lending to governments that face instabilities, either externally or internally created. Kind of like a central bank for the globe
- The World Bank usually engages in lending to developing countries to reduce poverty

World Economic Outlook April 2026

Class 20 Poll Everywhere

Learning objectives for Chapter 13

- When **shocks** hit aggregate demand or energy supplies and prices, governments often try to stabilize their economies
 - **Monetary policy** adjusts the short-term interest rate in a currency area like the U.S., the Euro Area, India, China, etc.
 - **Fiscal policy** adjusts spending G to directly change Y or adjusts taxes T and transfer payments Tr to indirectly change Y (Chapter 18)
- Adjusting policy is important to do in good times, when $\tilde{Y}$ and $\Delta\pi$ are too high, and in bad times, when $\tilde{Y}$ and $\Delta\pi$ are too low
- Central banks in advanced countries often set monetary policy to follow a **rule** based around an **inflation target** usually $\bar{\pi} = 0.02$
- By following the rule, central banks **build credibility** and anchor agents' inflation expectations, so that they actually can manage inflation

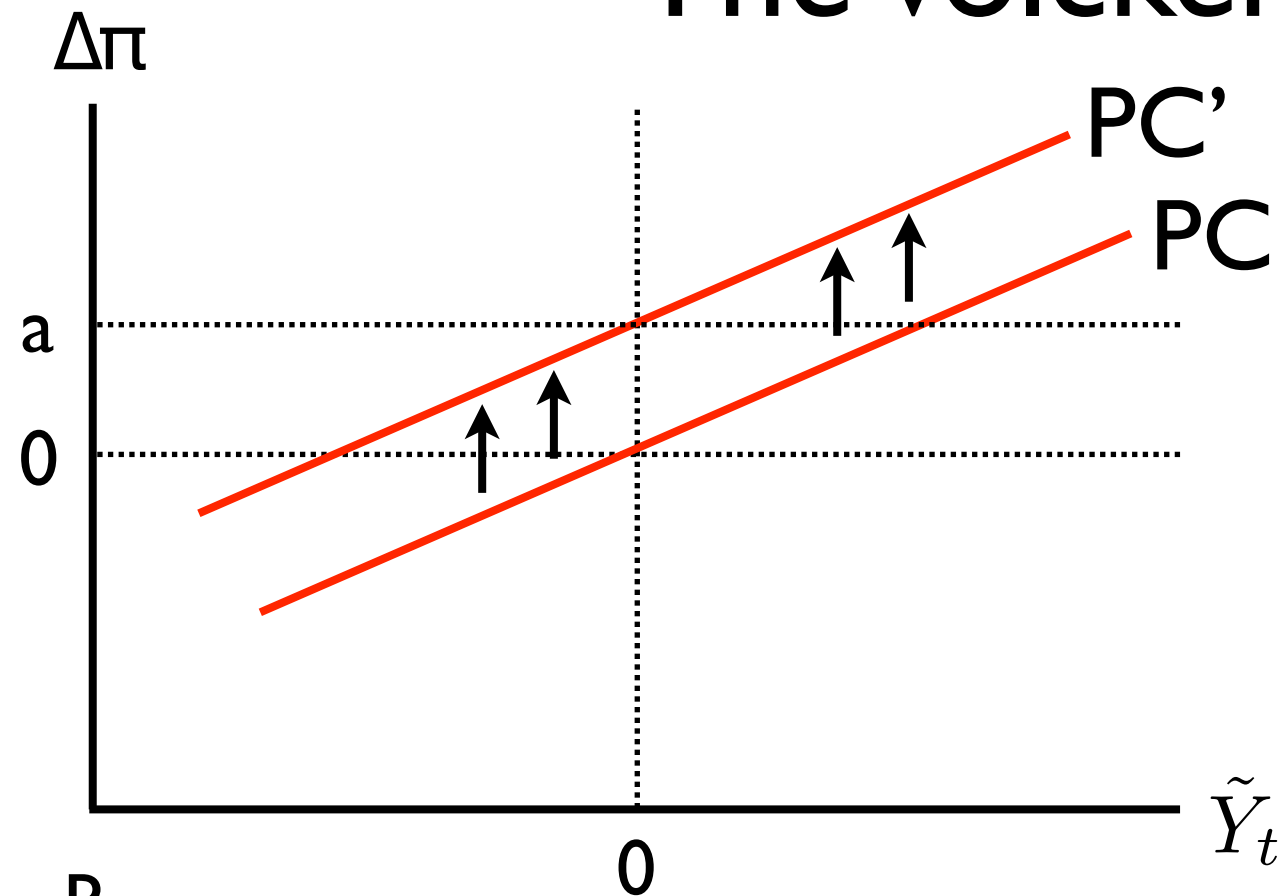
The Volcker Disinflation

- Policymakers agreed that high inflation required action
- [Paul Volcker](#) was nominated by President Jimmy Carter and [confirmed 98-0 by the Senate on August 2, 1979](#). The former chair, G. William Miller, became the Treasury Secretary. [History of chairs here](#)
- The August 1979 CPI release for July 1979 would have showed 11.5% inflation since July 1978
- People expected high inflation, having lived through years of it. Inflation was above 3% after September 1972
- The modern Phillips curve tells us

$$\pi_t = \pi_t^e + \bar{v}\tilde{Y}$$

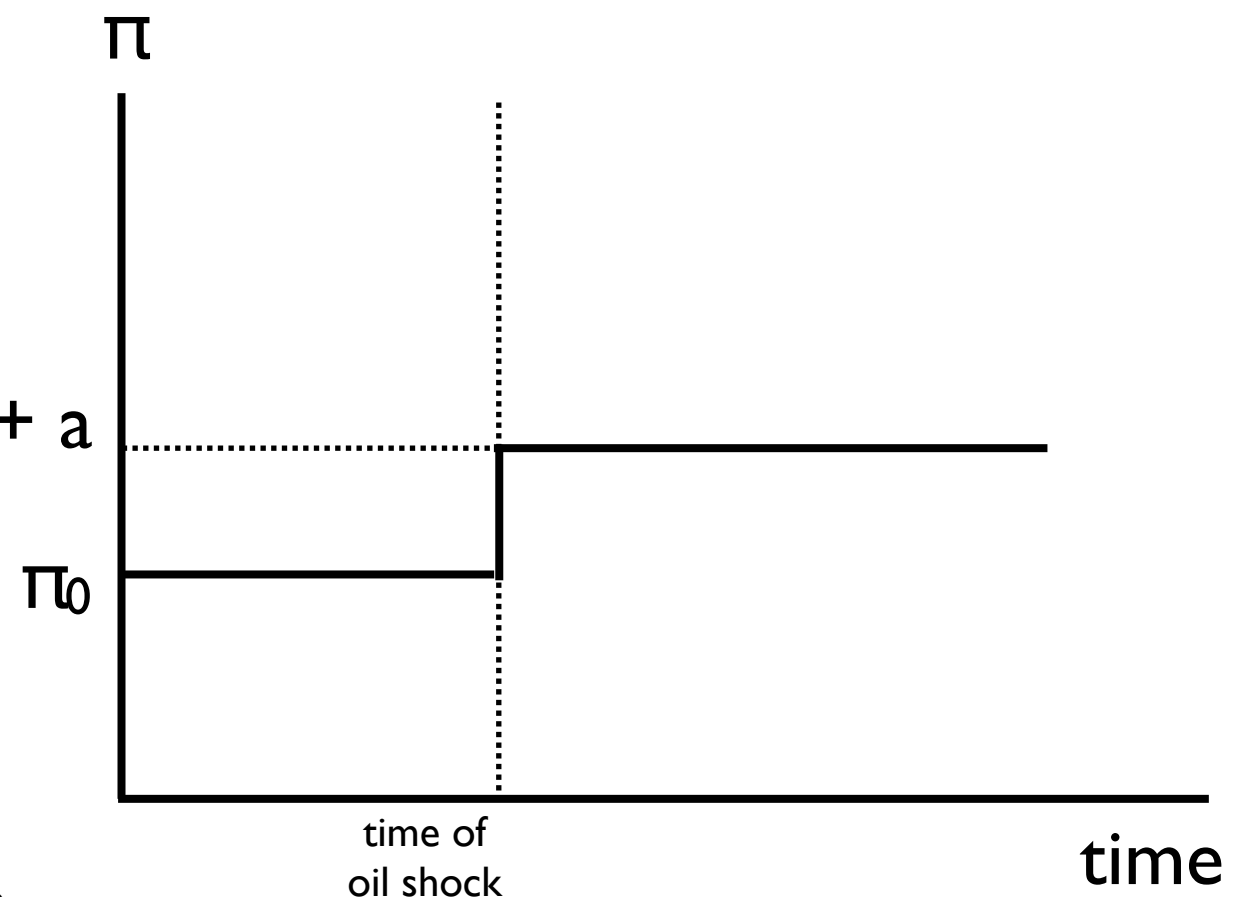
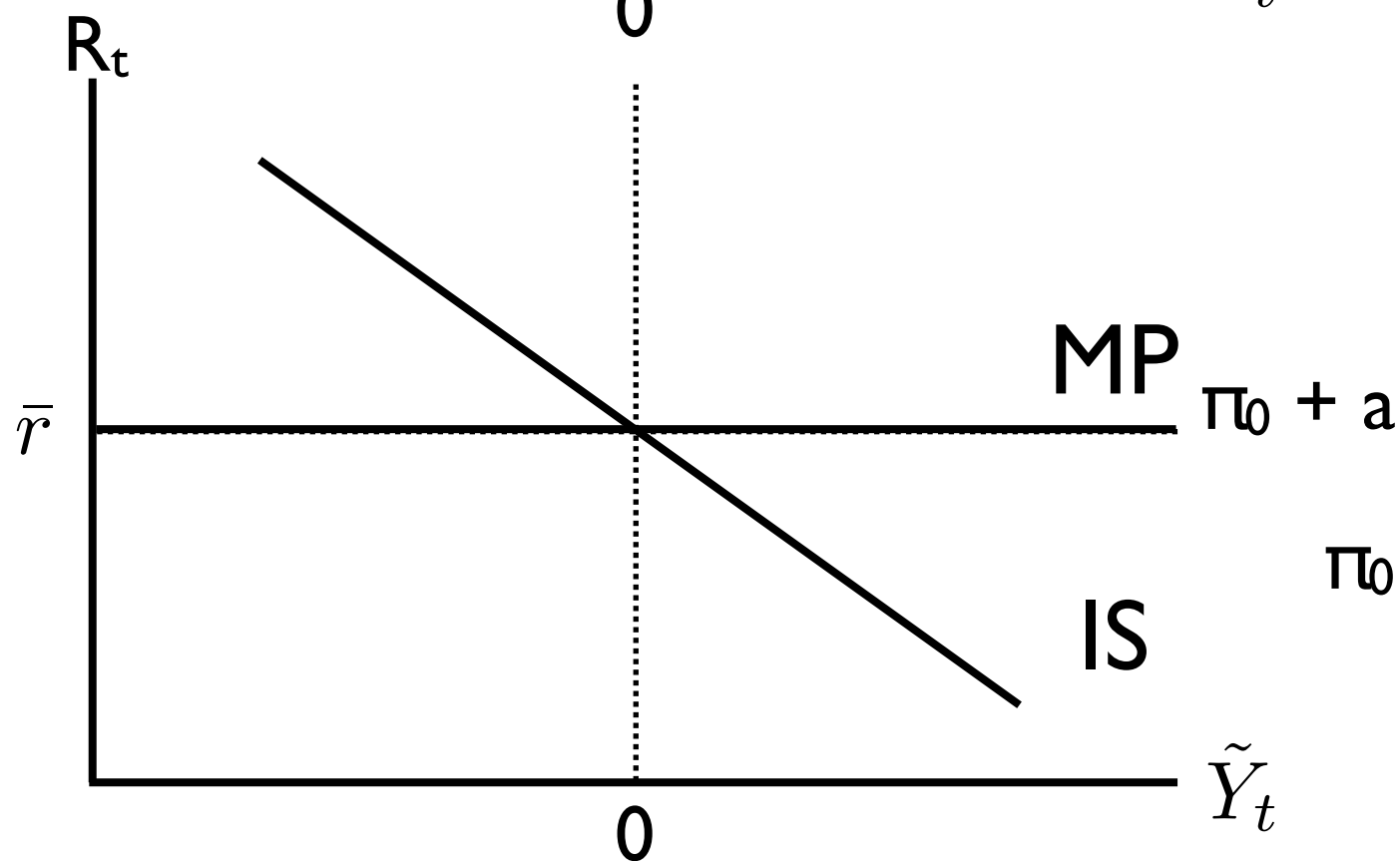
- When π^e is very high, the only thing the Fed can do is raise R high enough to push $\tilde{Y}$ very negative

The Volcker Disinflation

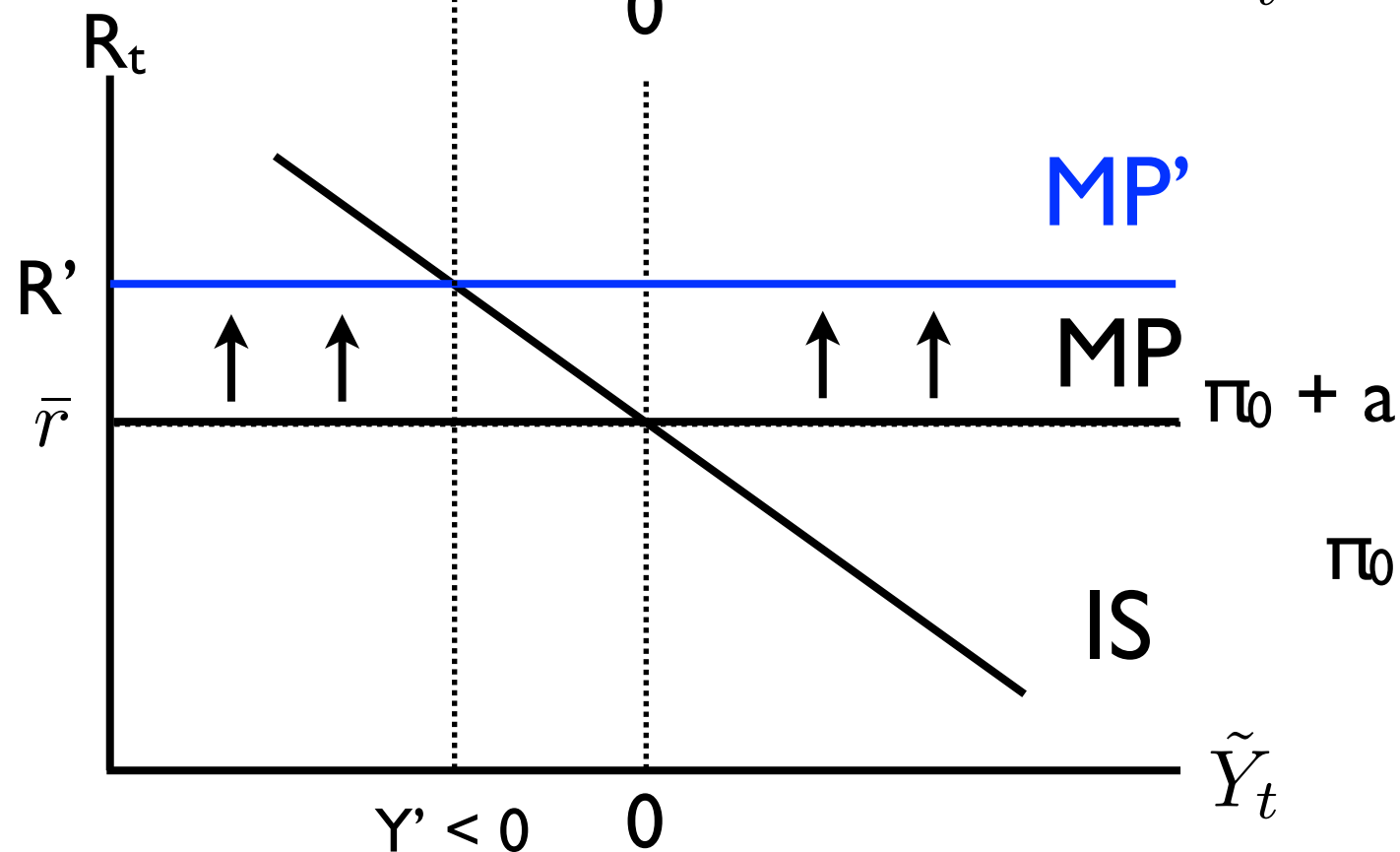
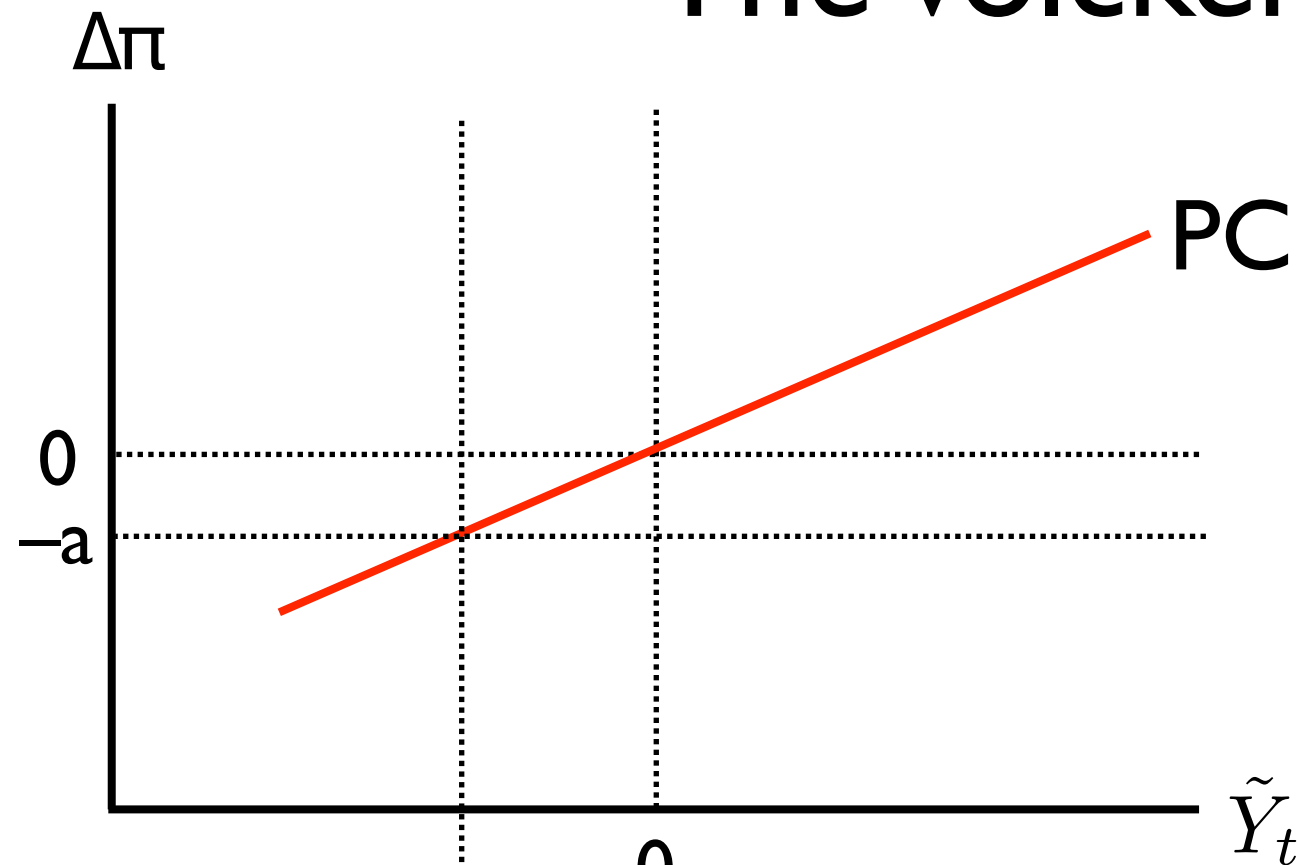


Q: Suppose there's an oil shock that increases all prices and inflation. What happens?

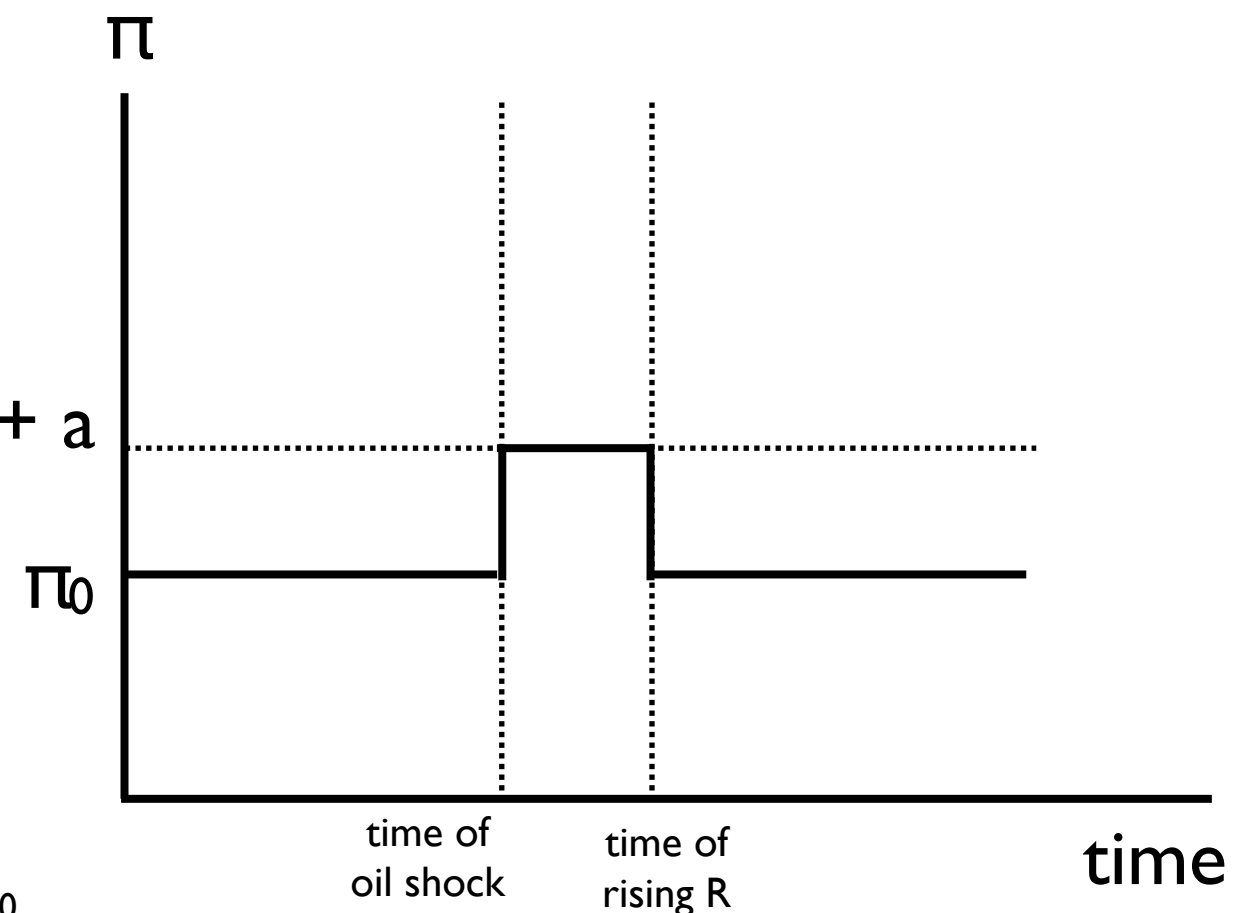
- The Phillips Curve PC jumps up to PC'
- Inflation increases to $\pi_0 + a$
- If nothing else happens, PC' will recede to PC
- And $\Delta\pi = 0$ again
- But the change in inflation would be permanent!



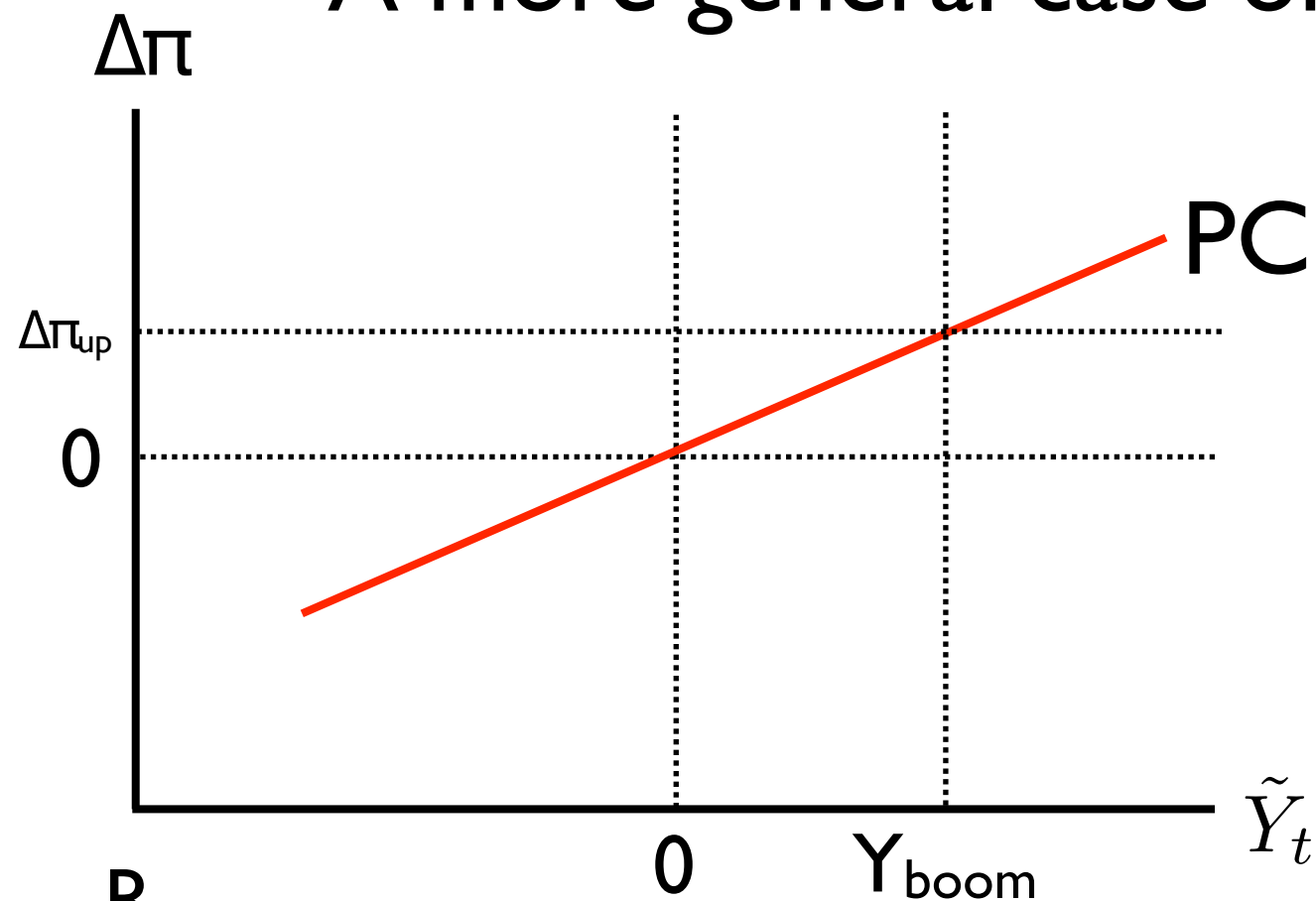
The Volcker Disinflation



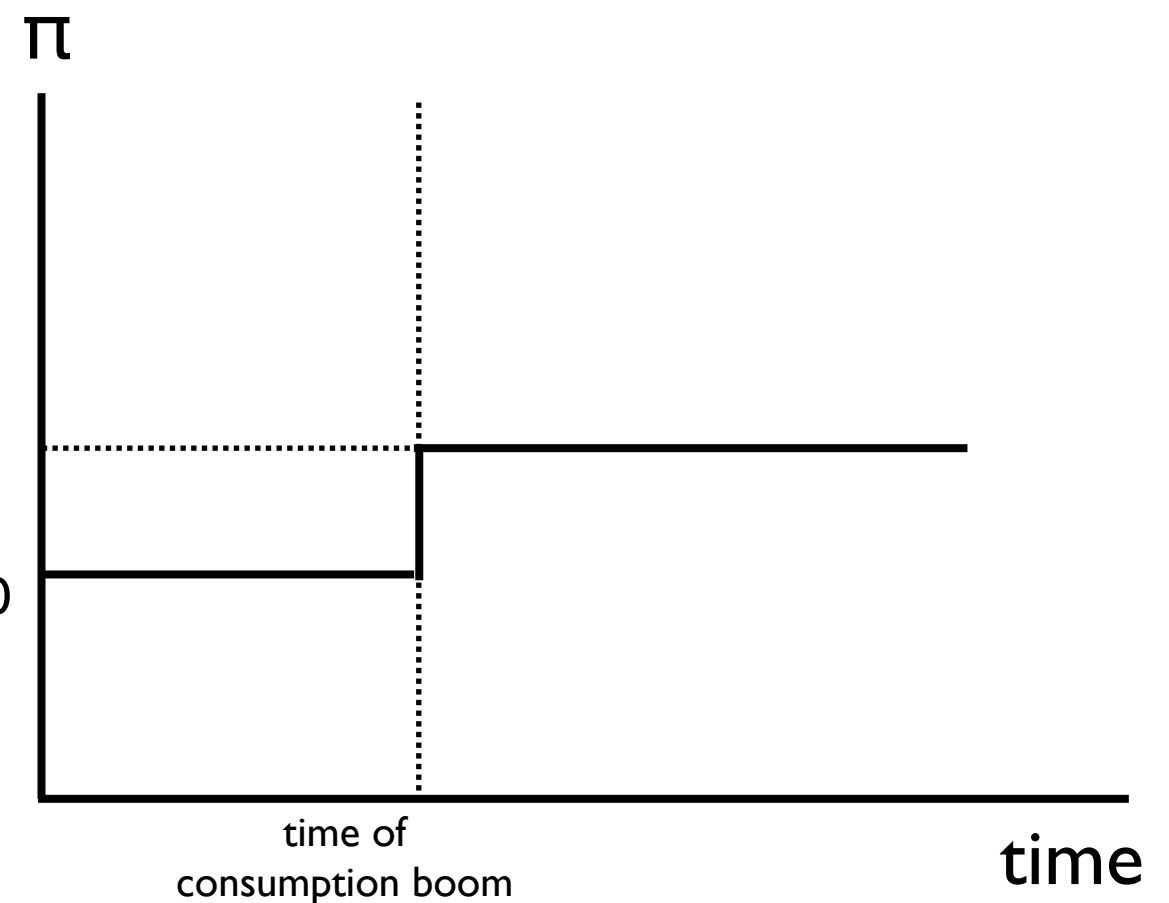
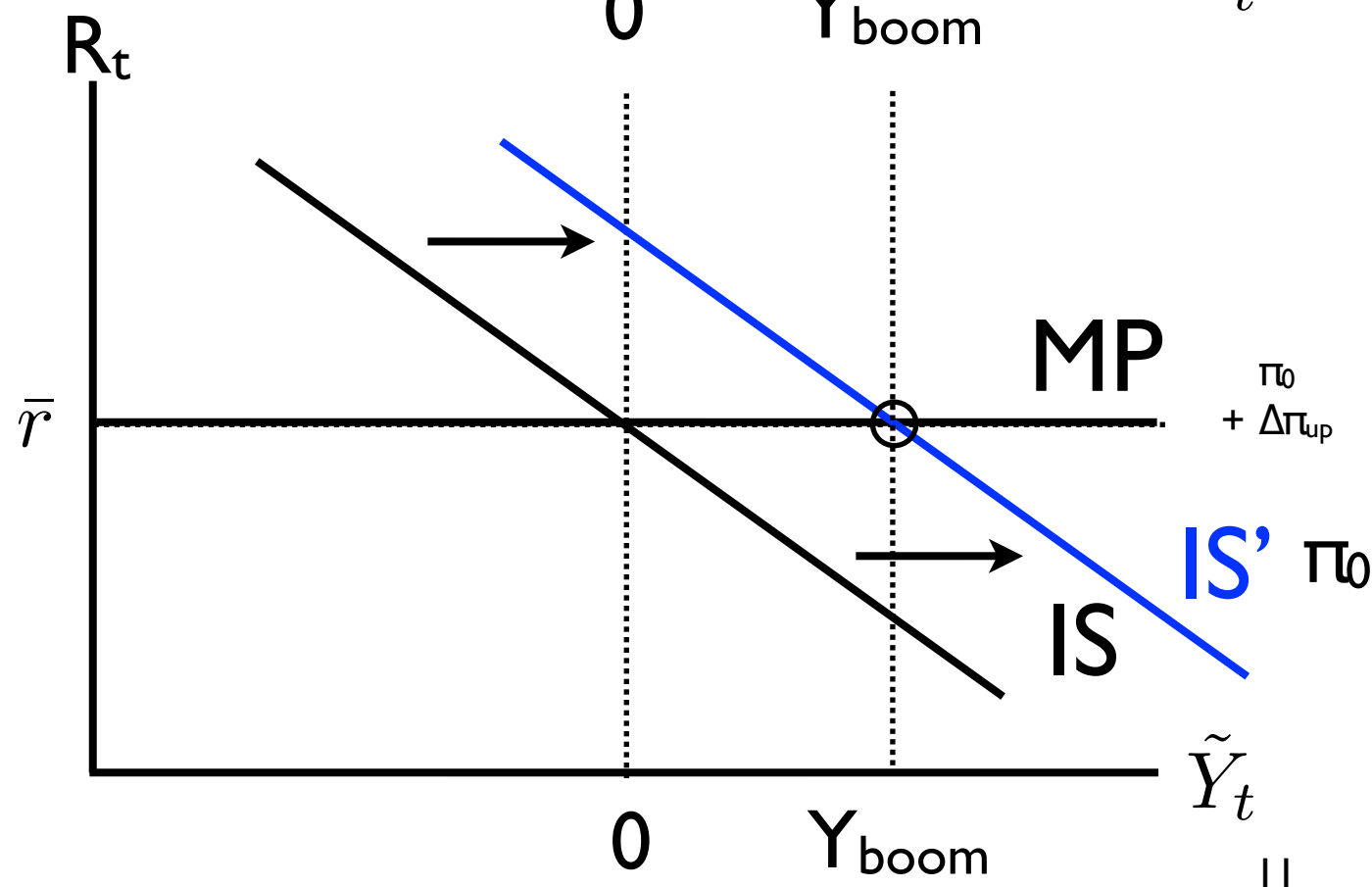
- Inflation rose because the Phillips Curve shifted
- The Fed must act to reduce inflation in the only way that it can: by causing a recession
- The Fed raises the real interest rate R by raising the MP Curve, causing a recession
- The recession then lowers inflation through the Phillips Curve
- Once inflation has fallen, the Fed lowers the real interest rate, and the recession ends



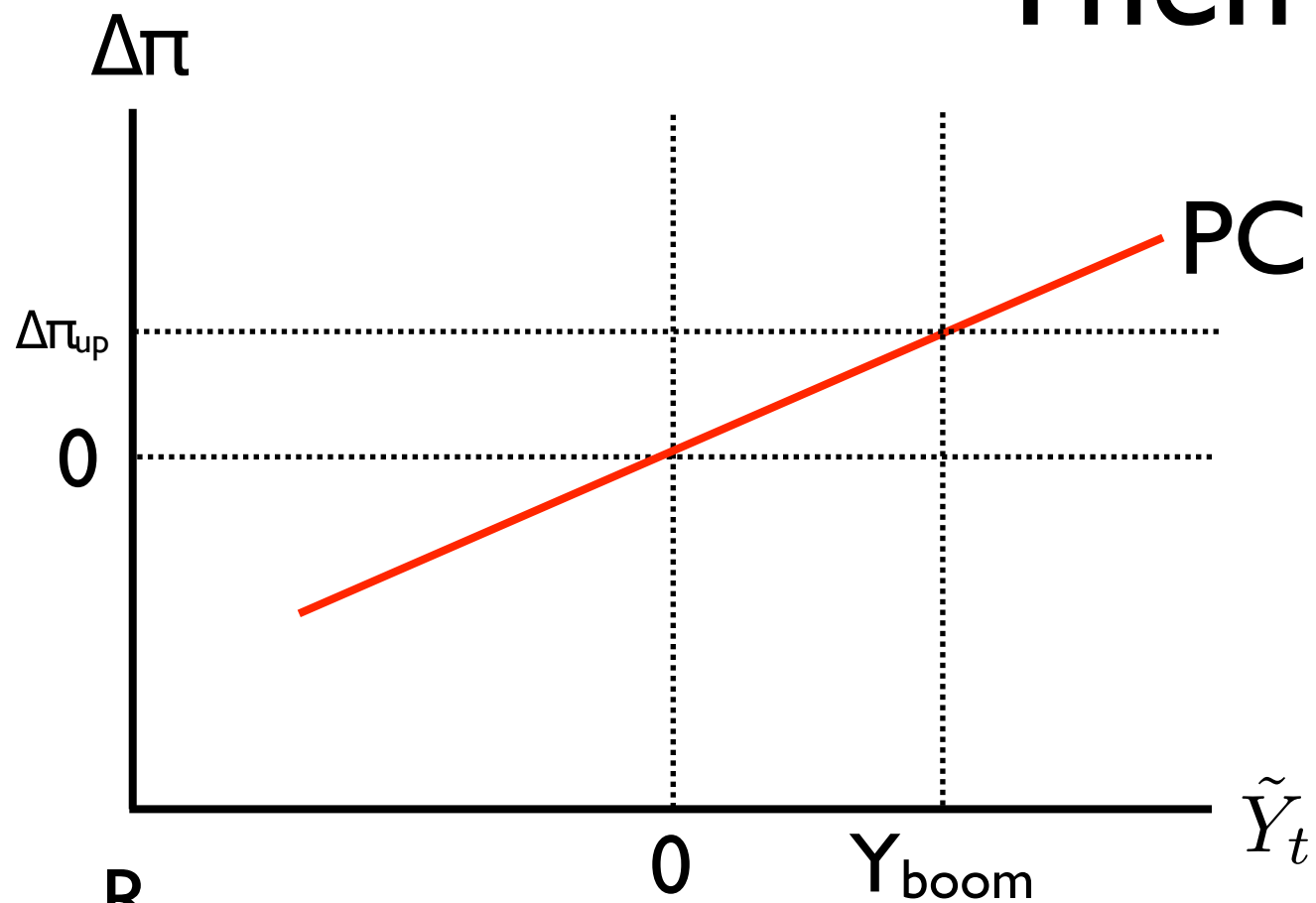
A more general case of “demand management”



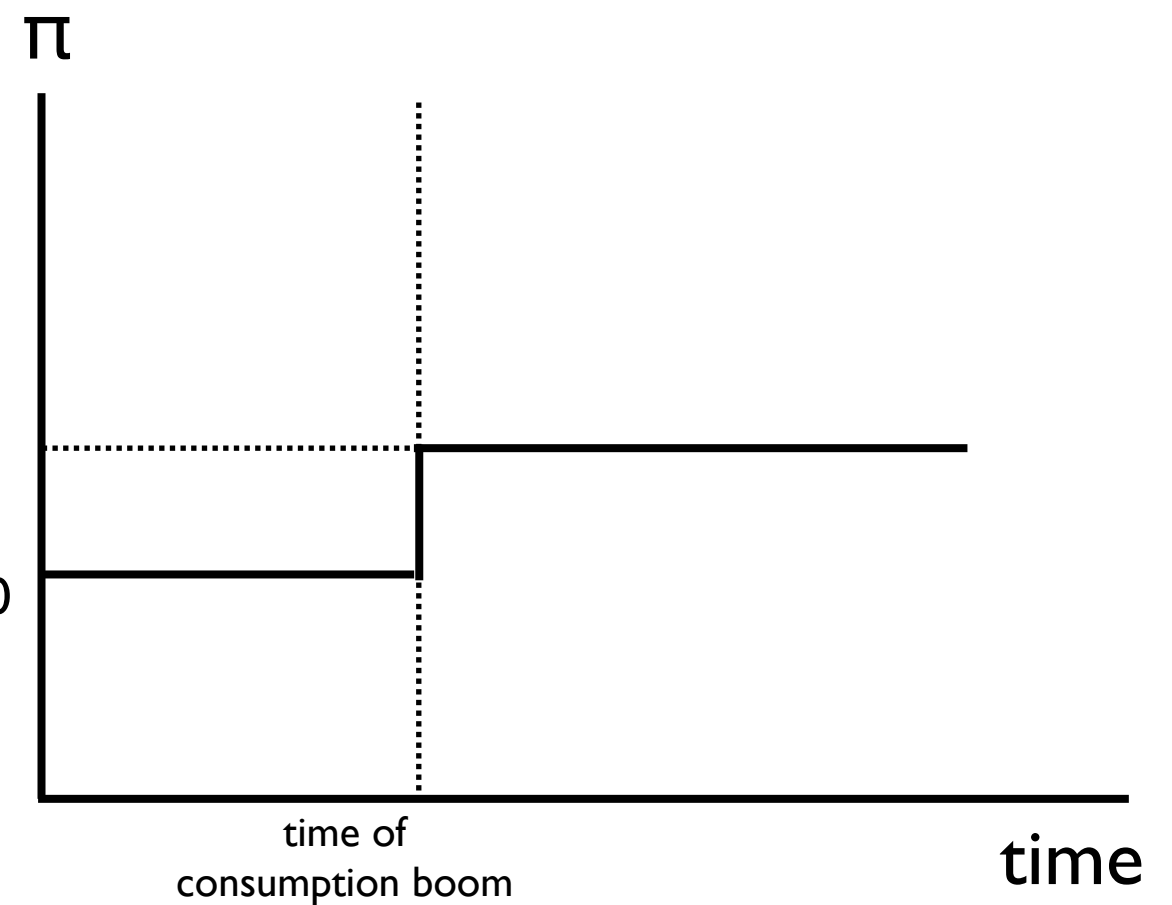
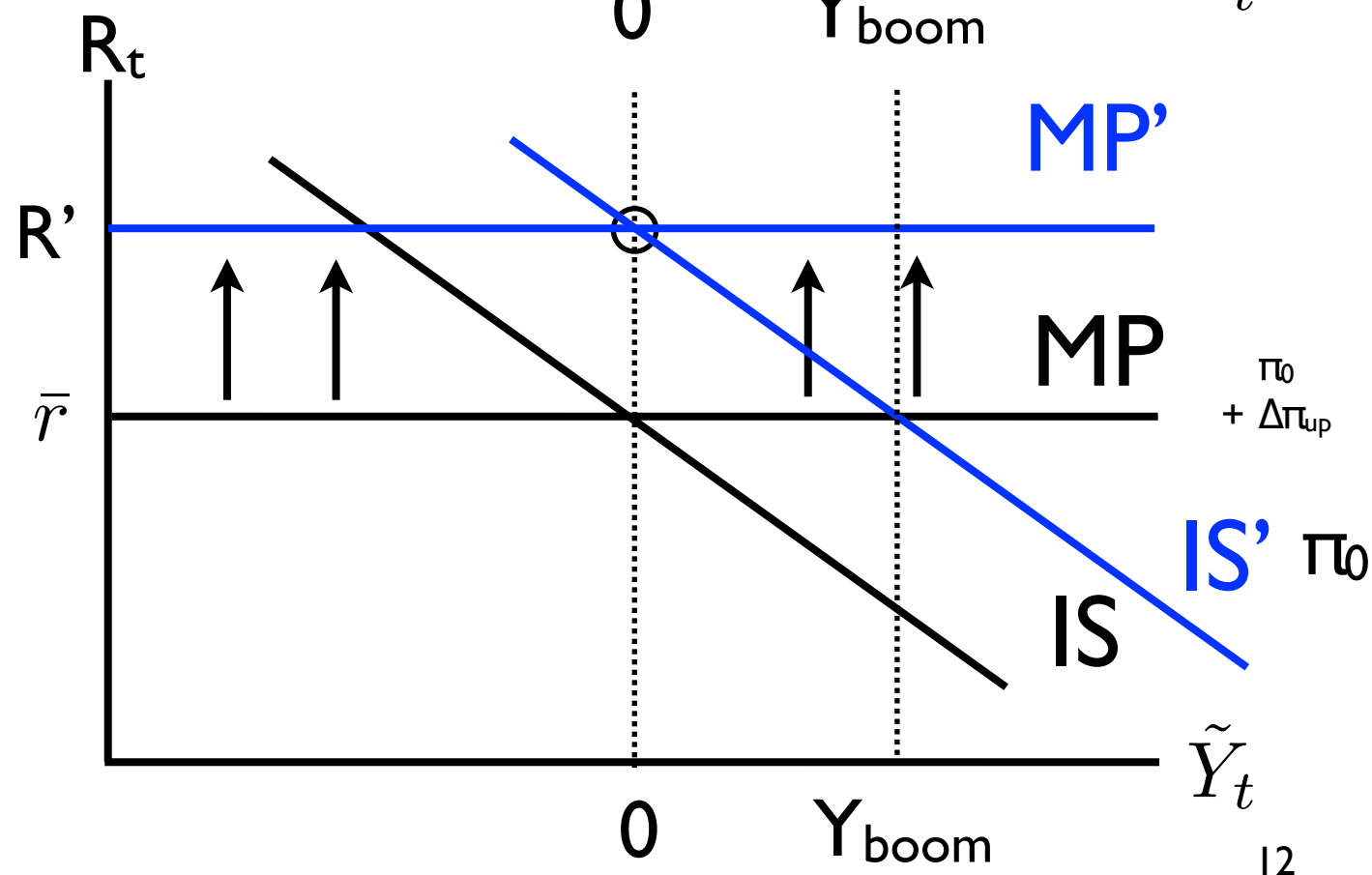
- There are many things that could shock the economy either negatively or positively
- Suppose U.S. consumers decide they want to consume a larger fraction of their permanent incomes. What happens in IS-MP / PC ?
- The IS Curve shifts out, because $\bar{a}_c = C / \bar{Y}$ rises, and the economy booms
- Through the Phillips Curve, inflation rises



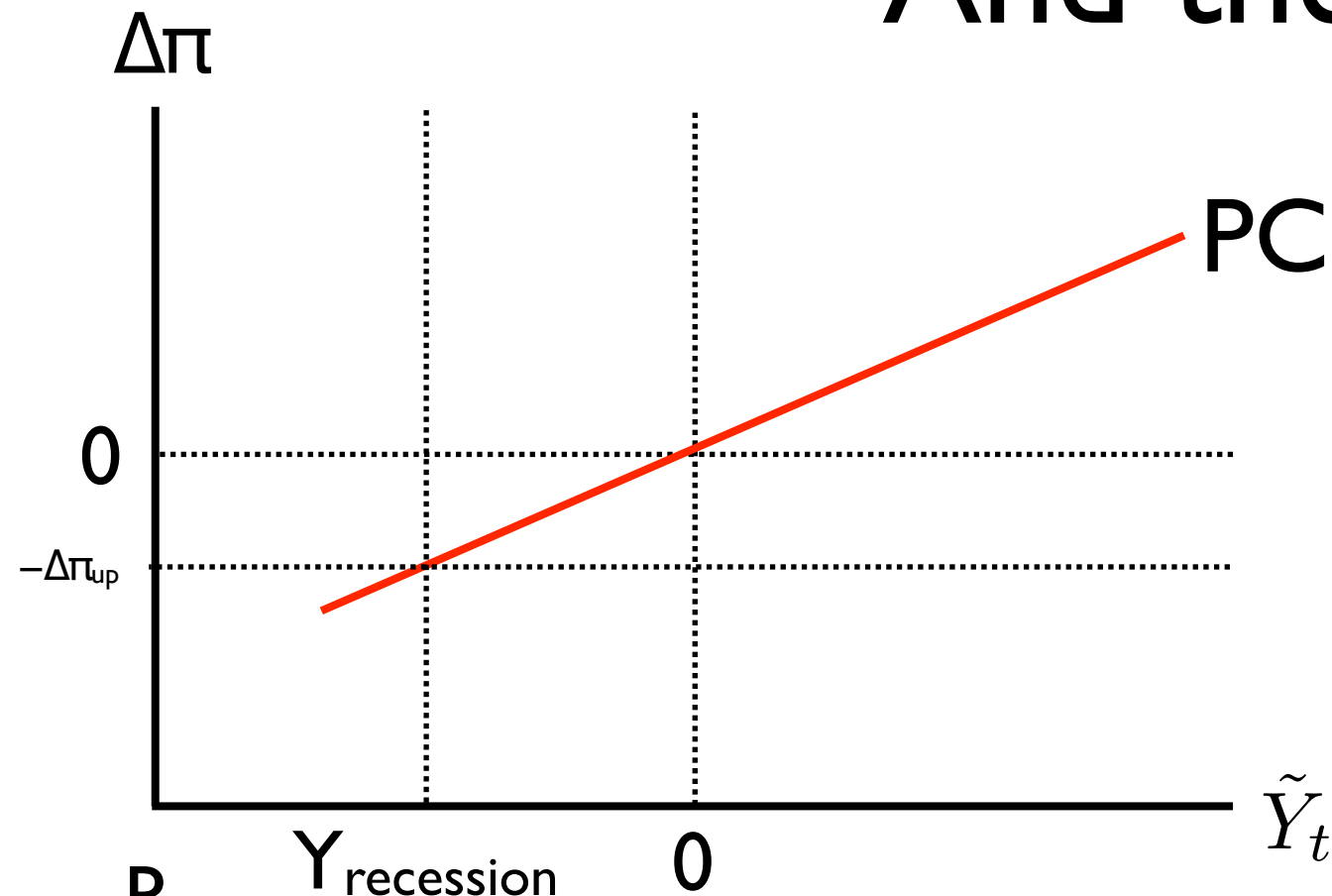
Then what?



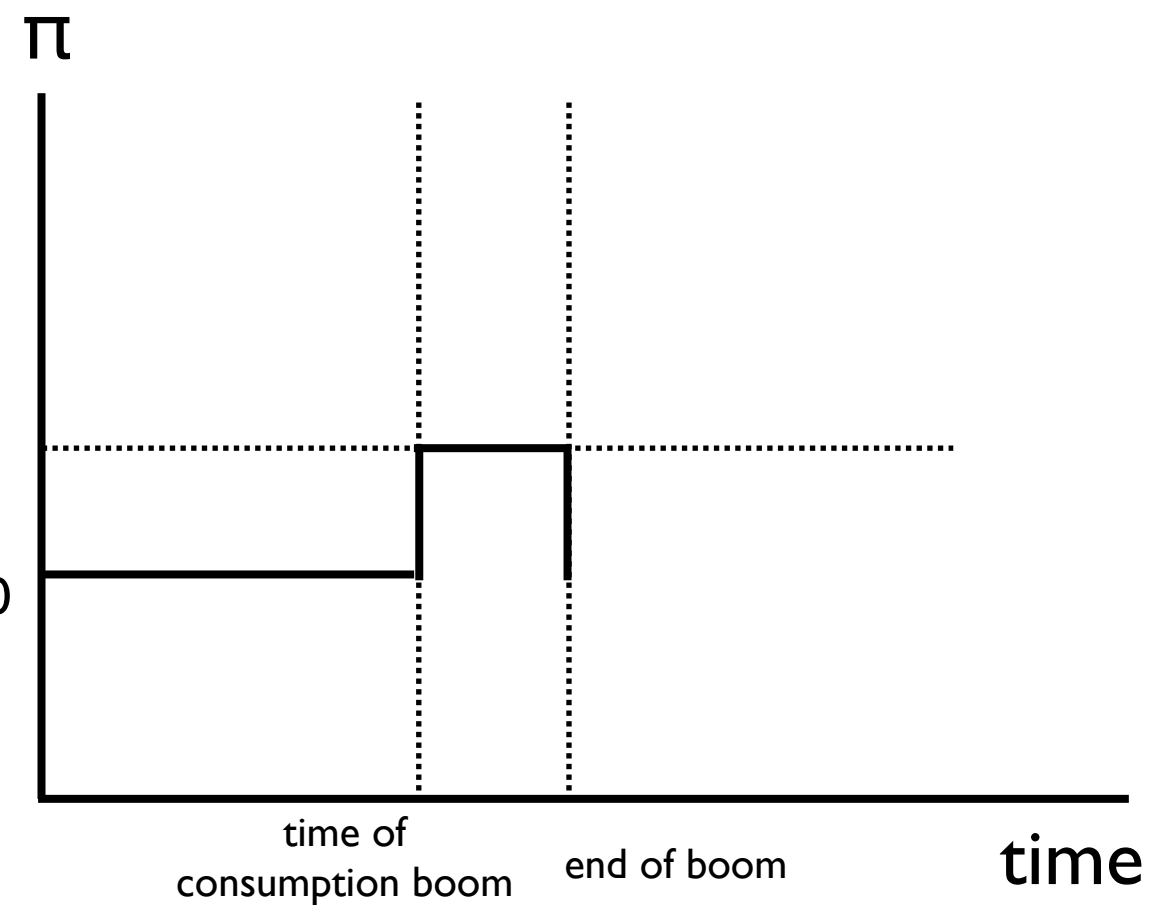
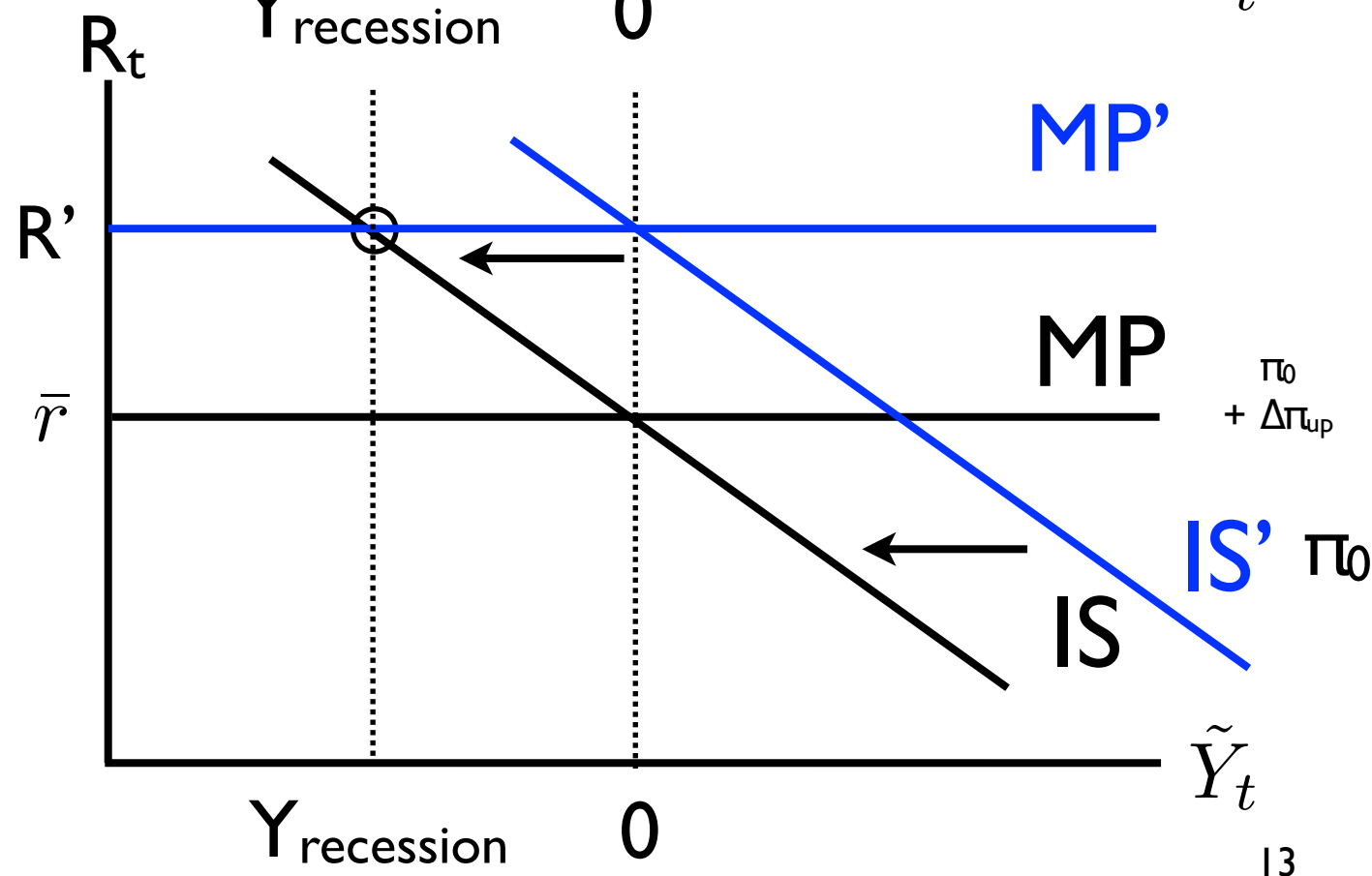
- In response to the rise in inflation above π_0 , the U.S. Federal Reserve will raise interest rates to combat inflation
- The MP Curve shifts upward
- If the increase is large enough, this will stop the rise in inflation



And then what?



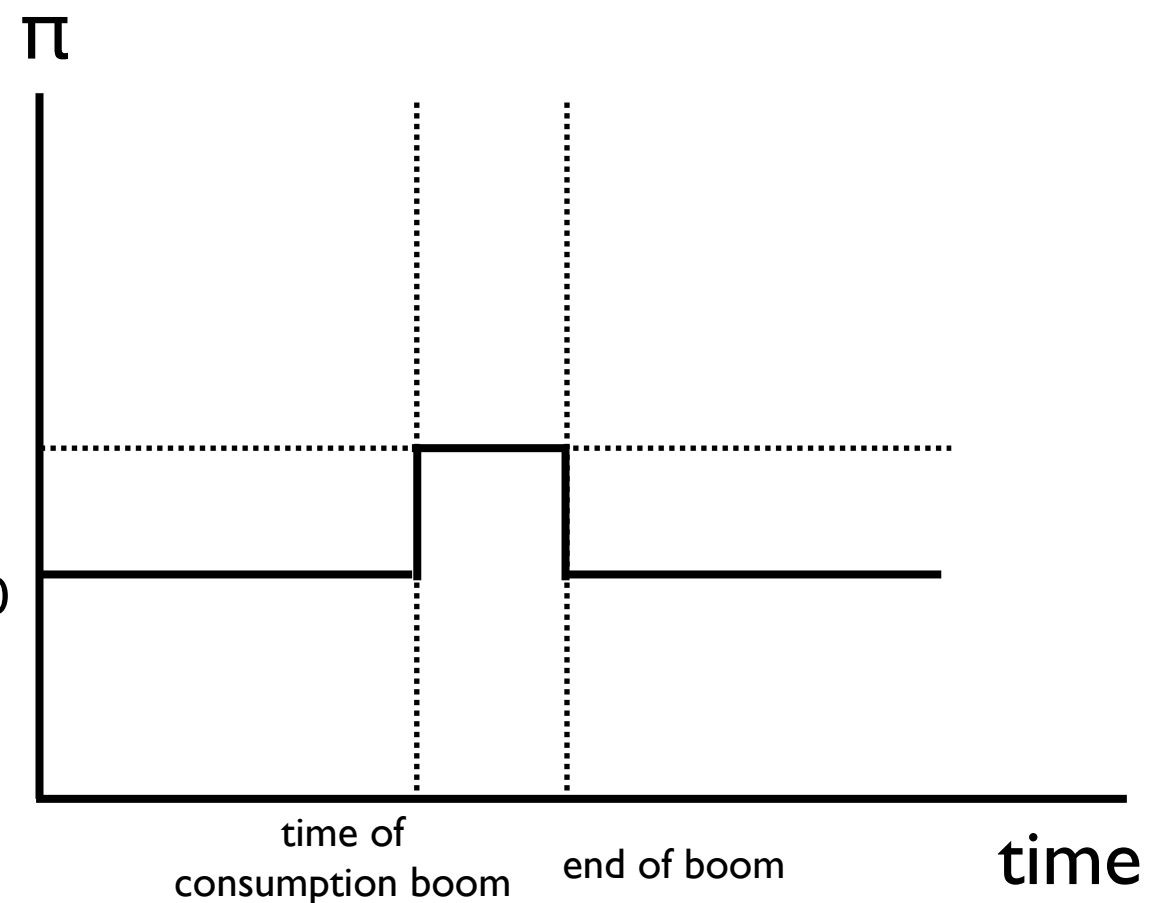
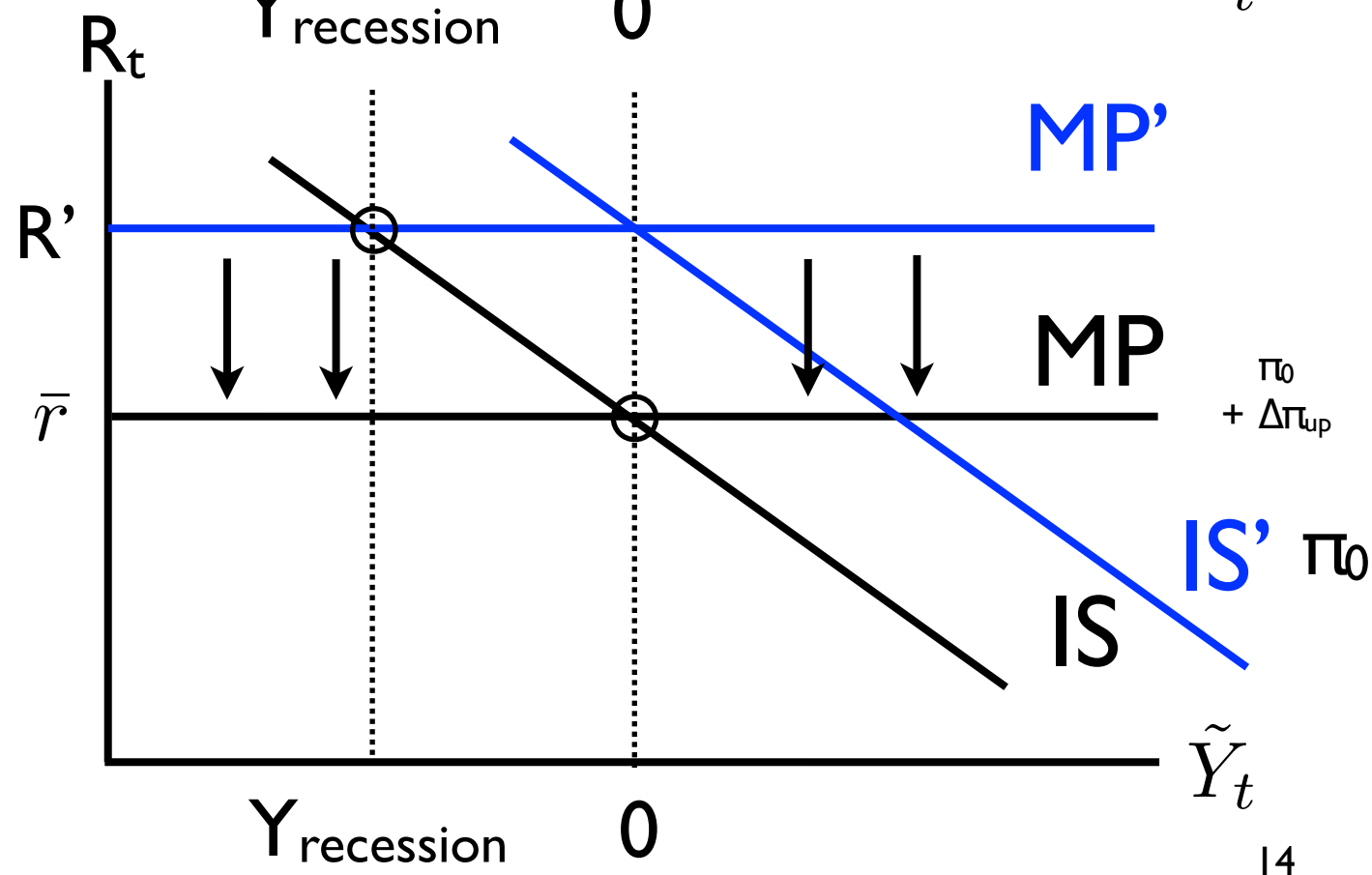
- Eventually, ***all*** positive shocks to aggregate demand must wear off; Consumption must return to its normal share of permanent income
- The IS Curve shifts back leftward, causing a recession because the interest rate is high
- This *lowers inflation* through the Phillips Curve



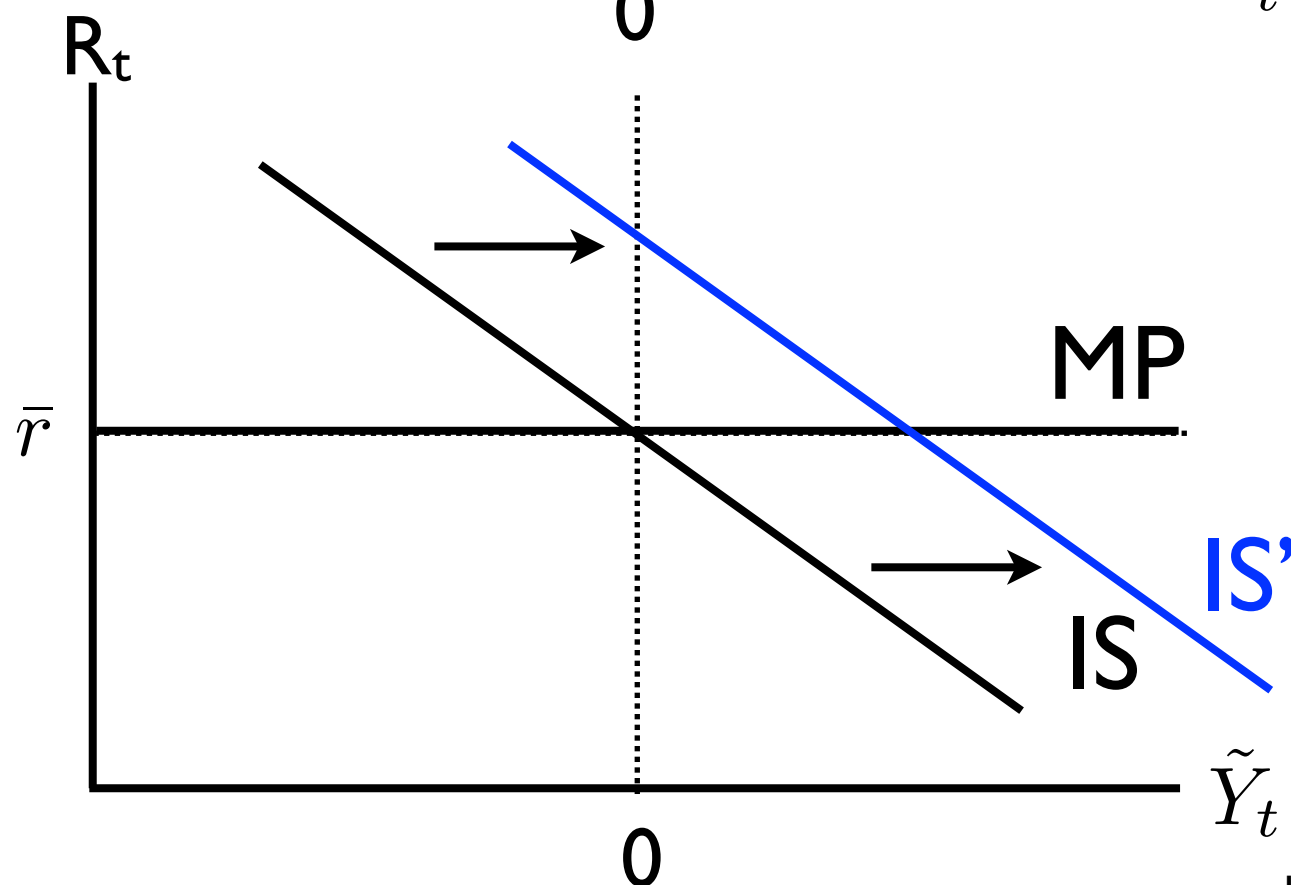
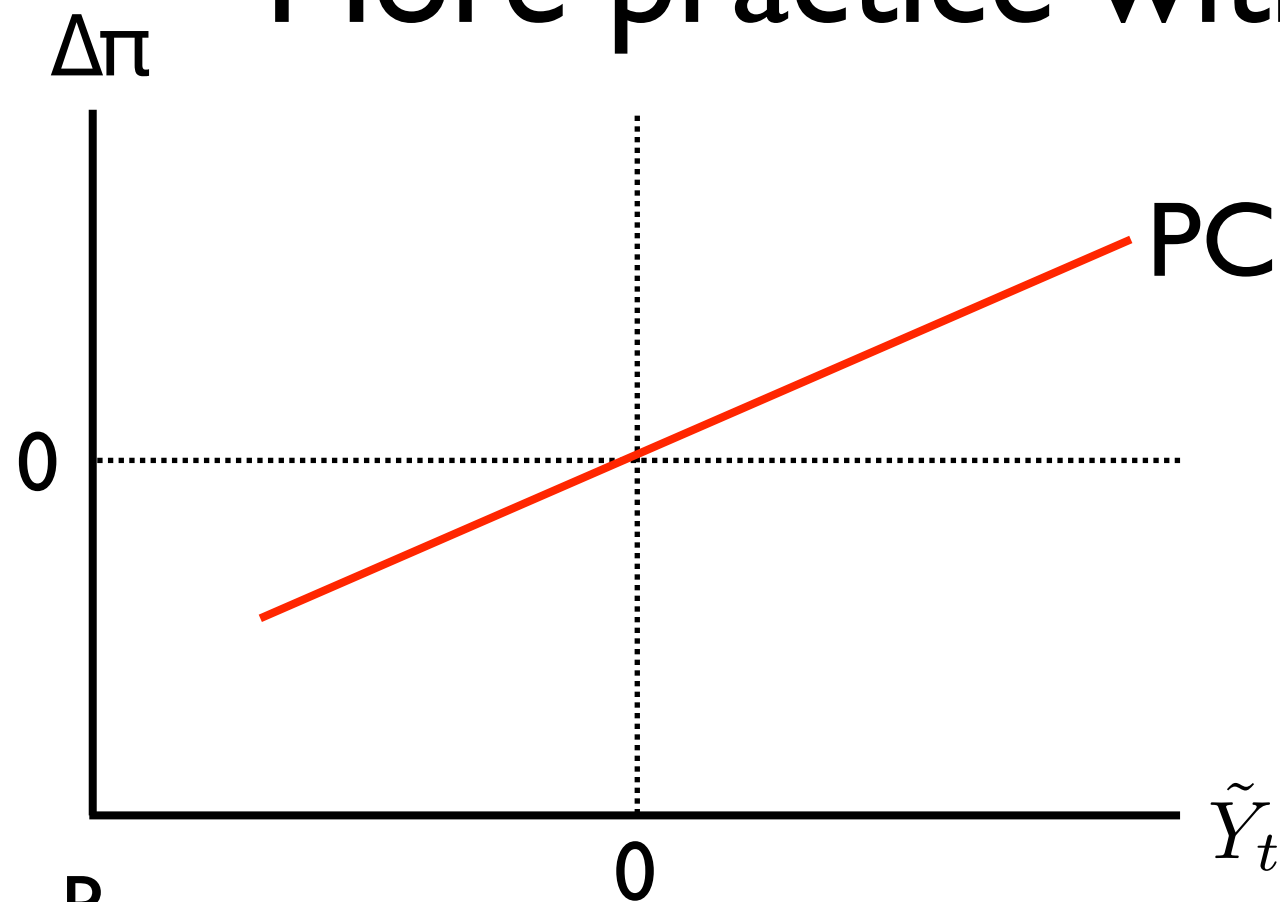
Then the Fed can lower interest rates



- Once the shock has worn off and inflation has returned to its normal level, the Fed lowers interest rates back to normal
- The MP Curve falls, and we're back where we started, at short-run output equal to zero, neither a boom nor a bust
- Through the Phillips Curve, inflation is stable



More practice with IS-MP & PC (1/3)

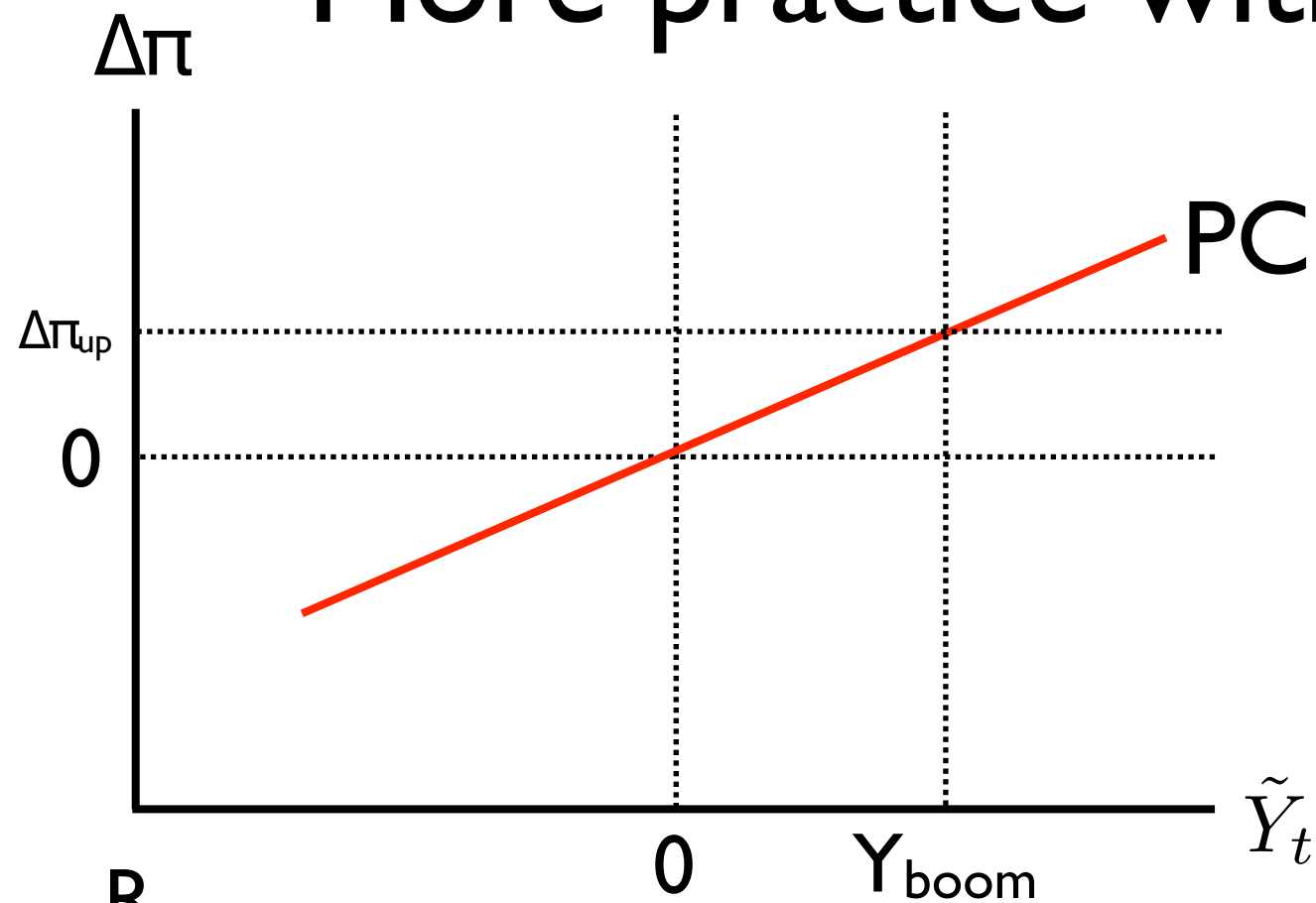


Q: Suppose the government expands its spending temporarily in order to stimulate the economy. What happens in the IS-MP or PC diagrams?

$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$

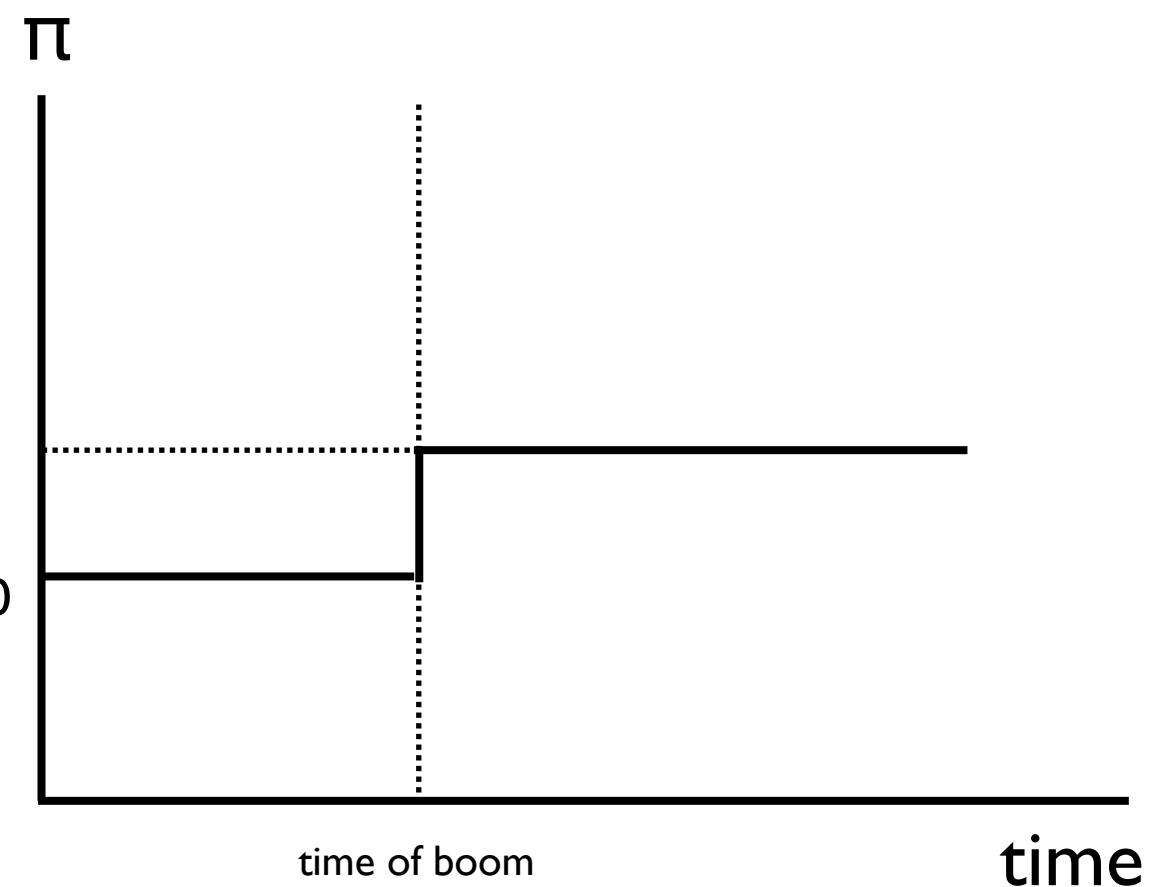
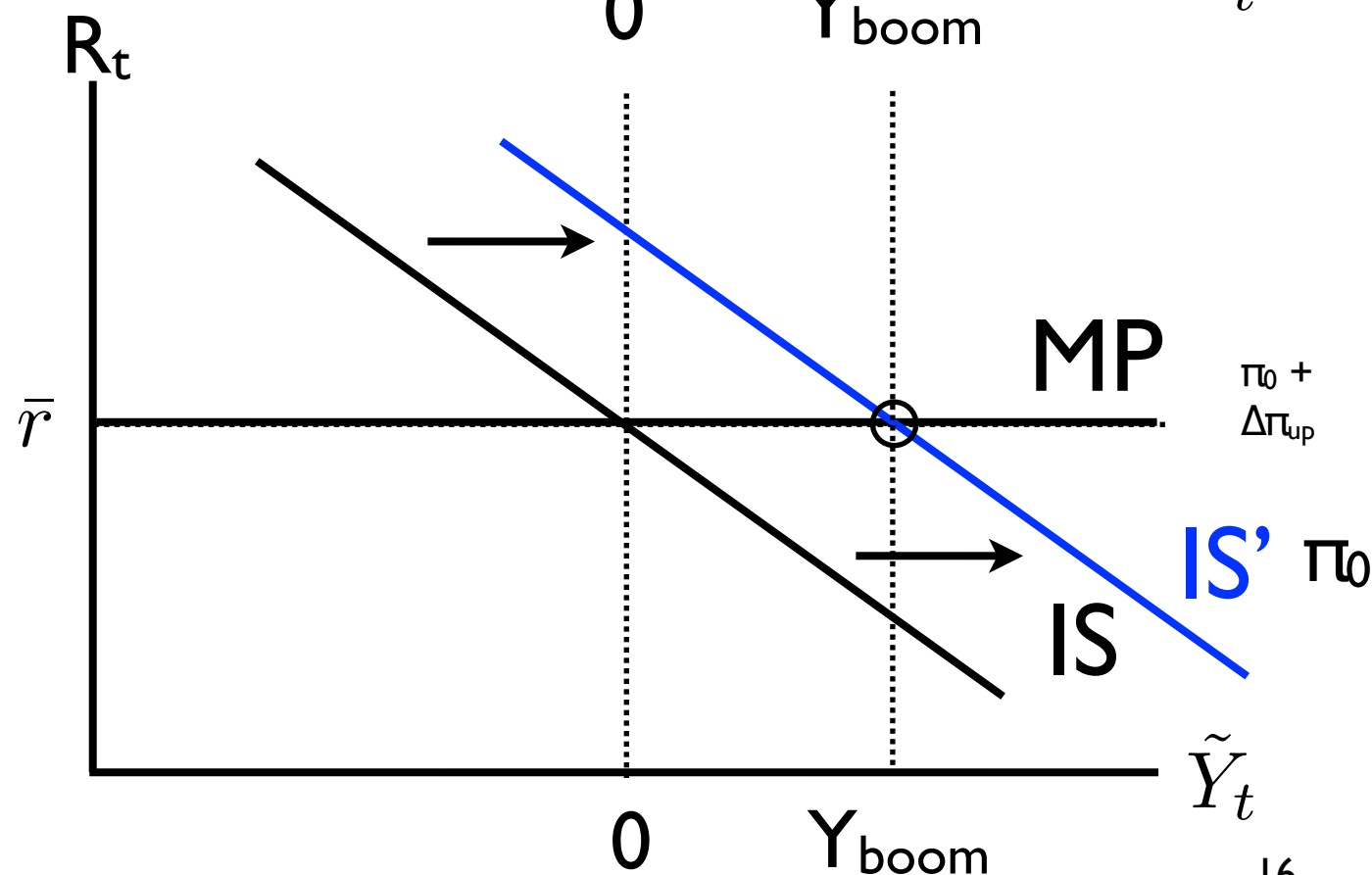
- The government temporarily spends more than its usual share of permanent income $\bar{a}_G = 0.20$
- When $\bar{a}_G$ rises, that pulls $\bar{a}$, the intercept of the IS curve, up with it.
- This is a *positive* aggregate demand shock of an amount equal to the increase in $\bar{a}_G$
- It's a **rightward shift** in the IS curve to IS'

More practice with IS-MP & PC (2/3)

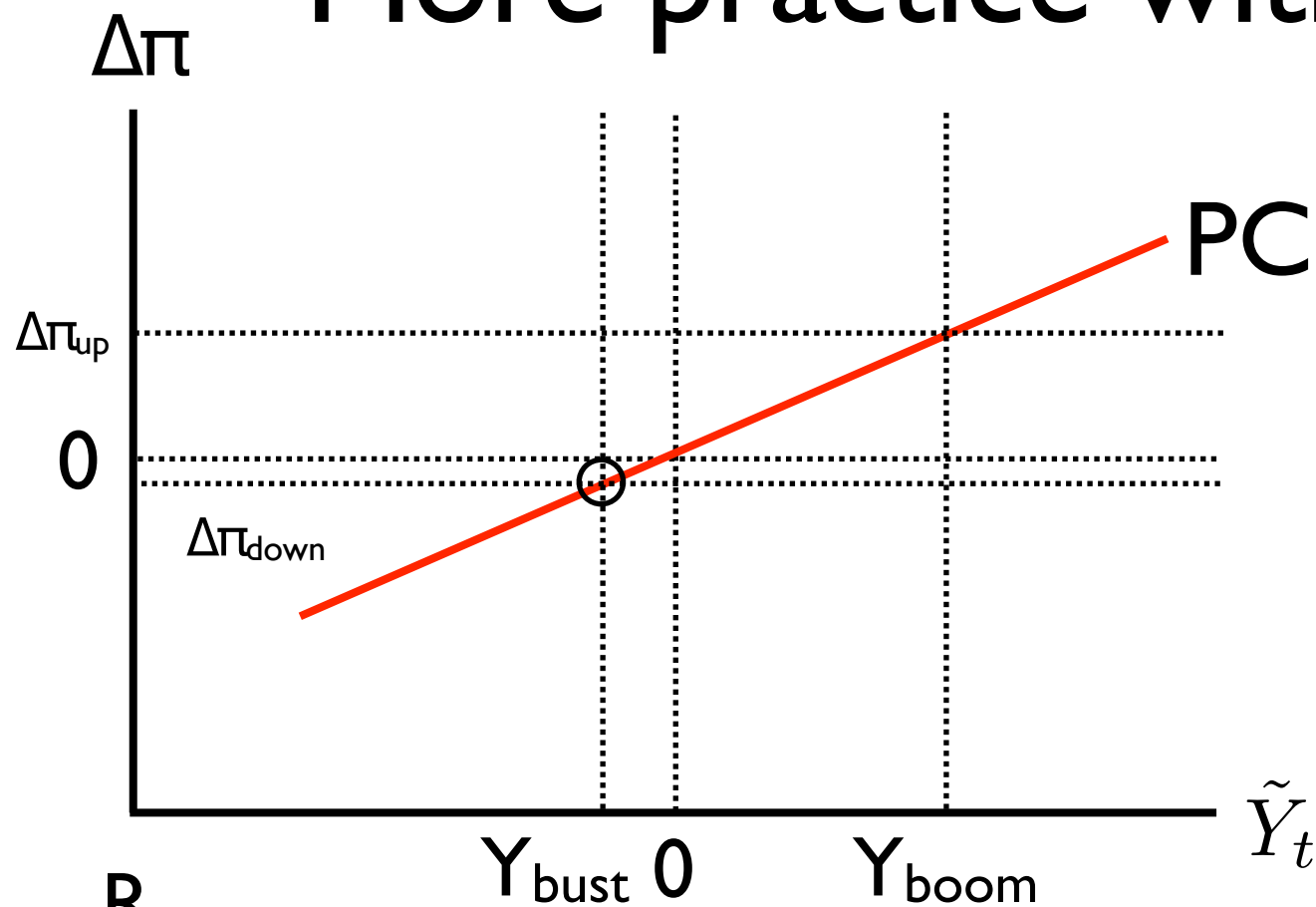


Q: Suppose the government expands its spending temporarily in order to stimulate the economy. What happens in the IS-MP or PC diagrams?

- The IS Curve has shifted to the right
- This produces a boom!
- This means that unless the Fed raises the real interest rate R , inflation will increase!

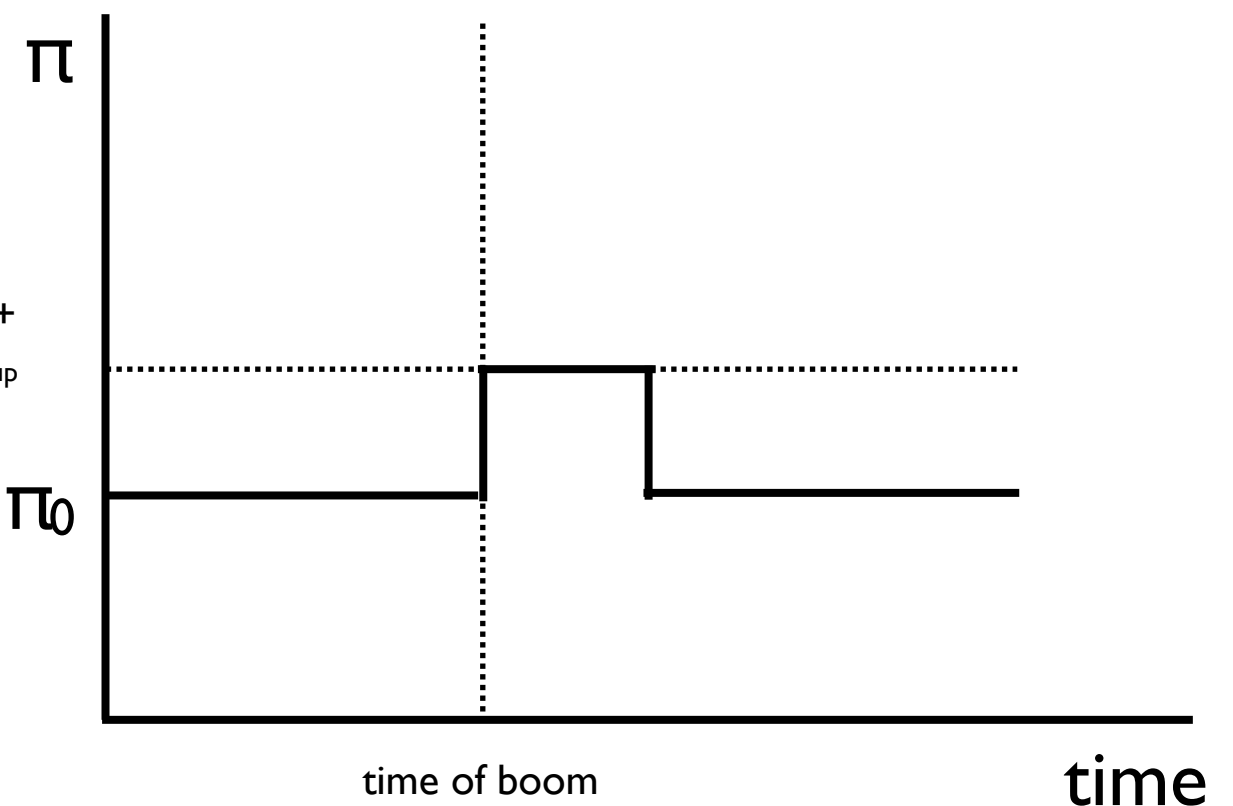
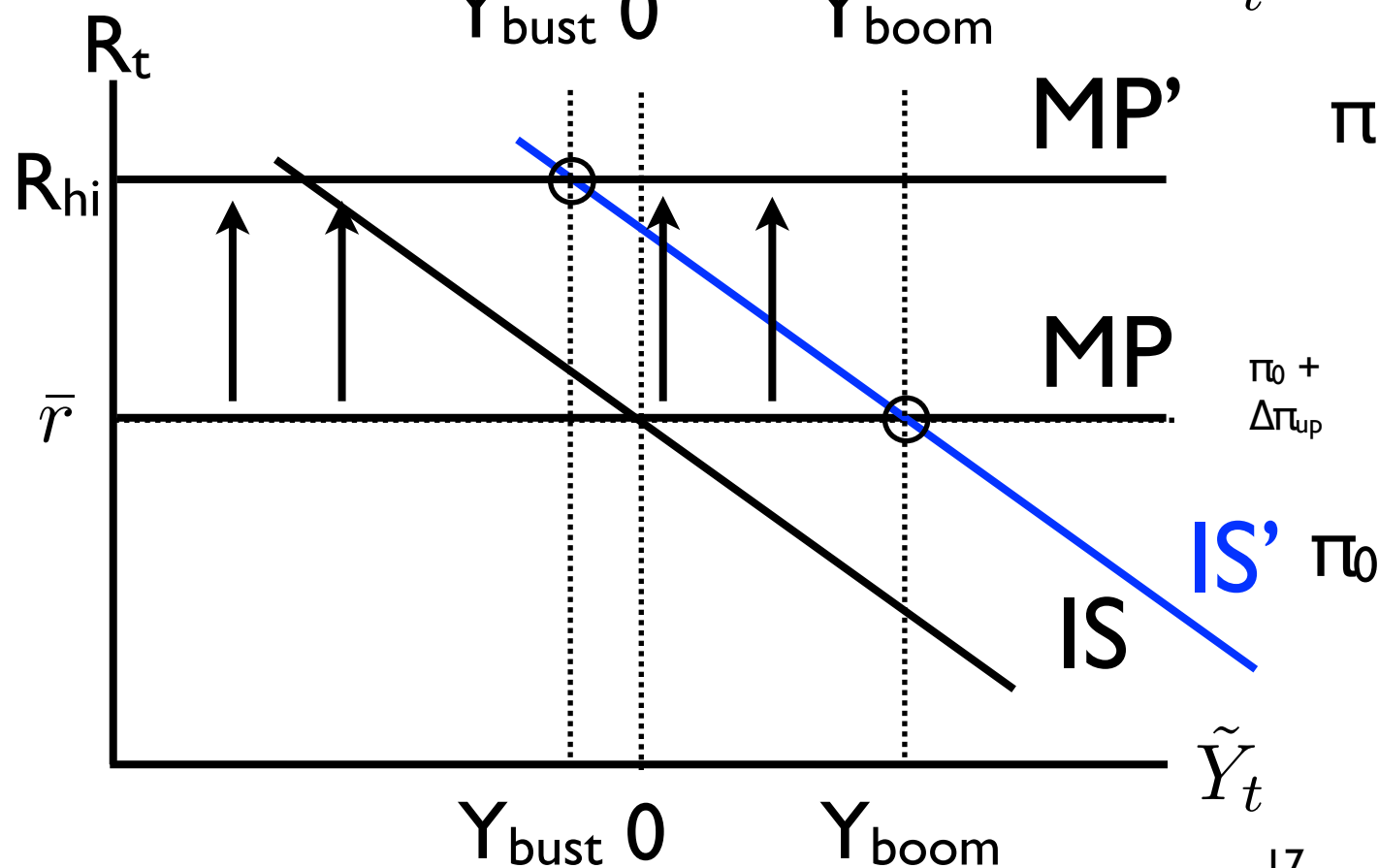


More practice with IS-MP & PC (3/3)

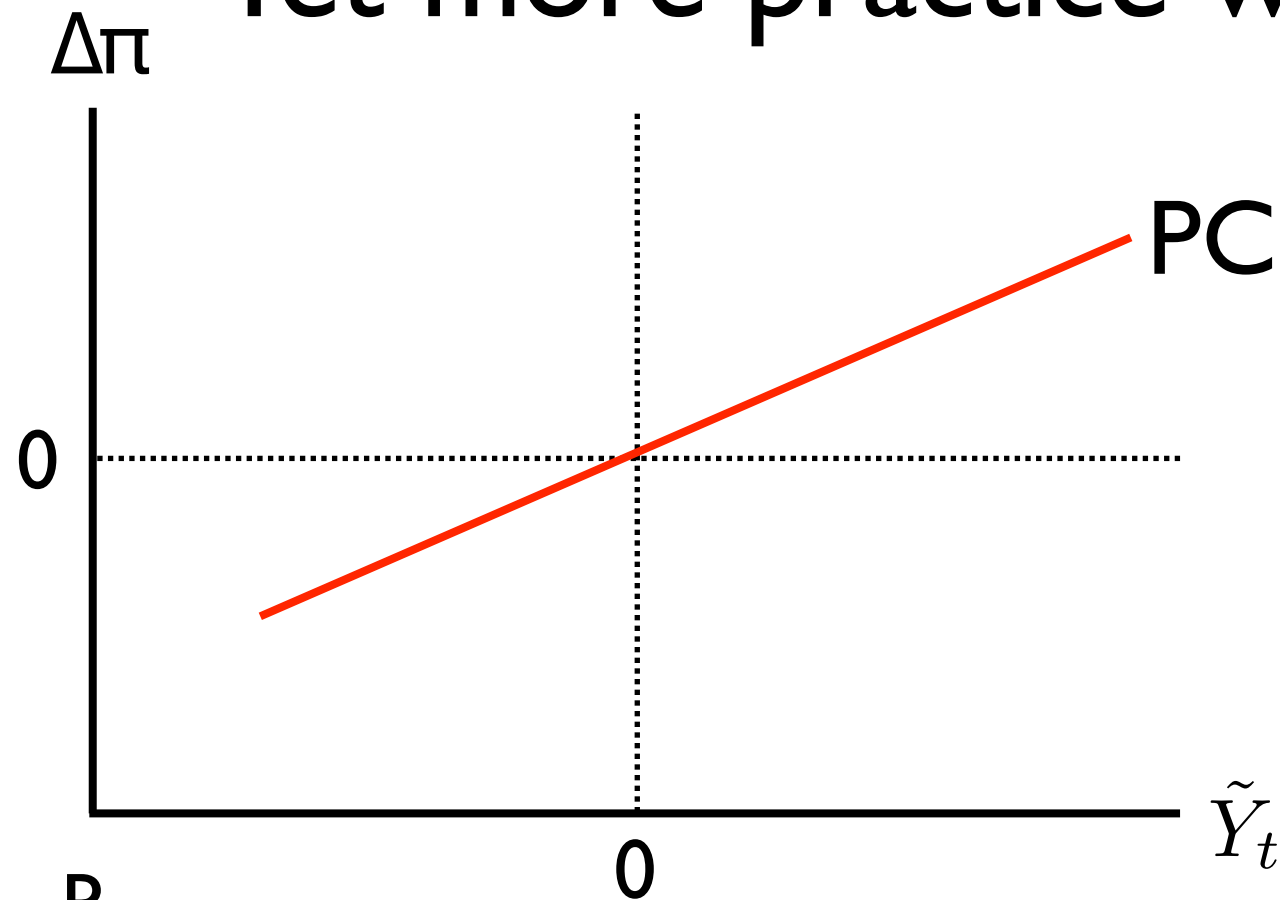


Q: Suppose the government expands its spending temporarily in order to stimulate the economy. What happens in the IS-MP or PC diagrams?

- What should the Fed do if it wants to pull inflation back down toward π_0 ?
- Raise the interest rate R and cause a recession
- Inflation will fall, and if the Fed keeps R high sufficiently long, π falls all the way back to π_0

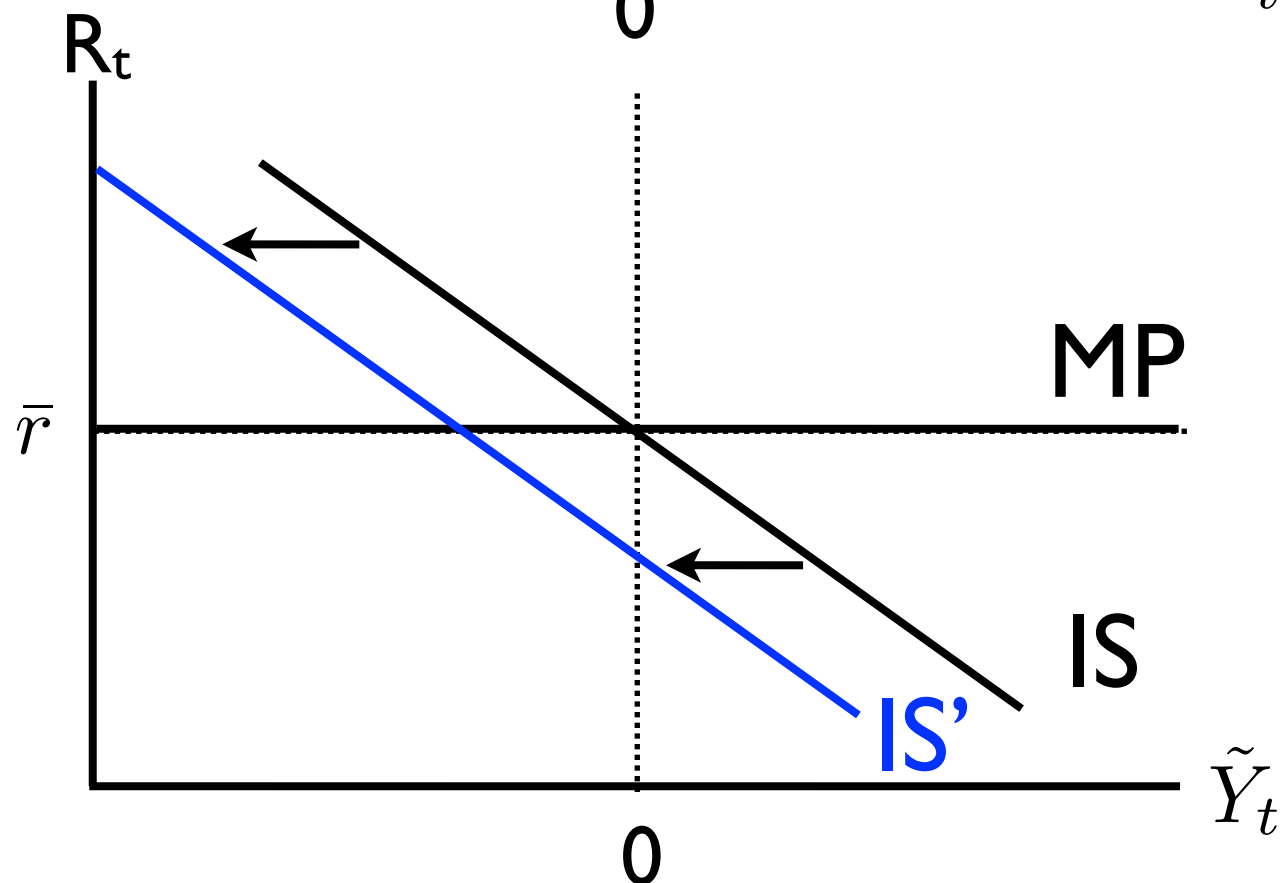


Yet more practice with IS-MP & PC (1/3)



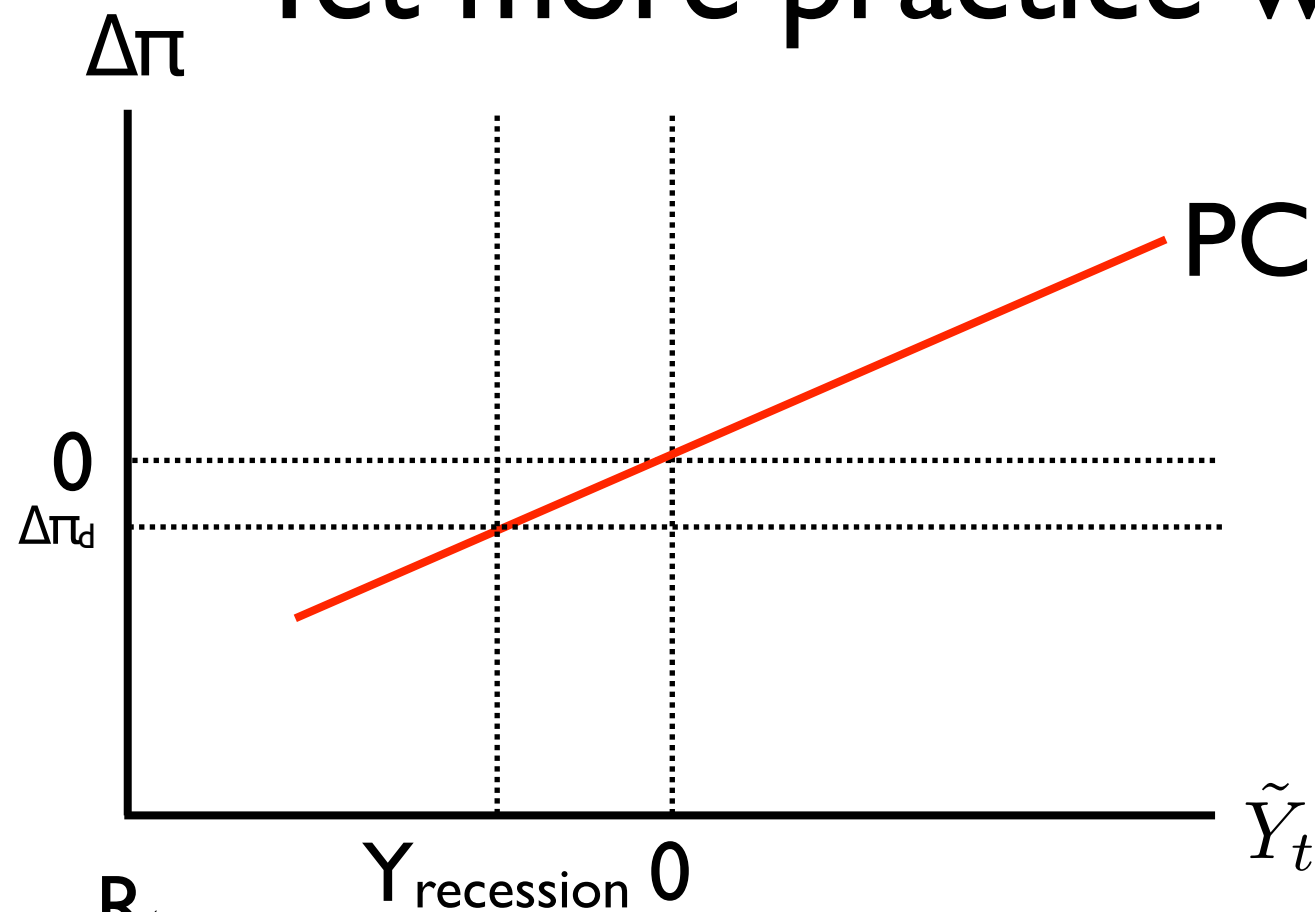
Q: Suppose consumers feel their houses, a major part of their wealth, have lost a lot of value, so they temporarily cut back on spending. What happens in the IS-MP or PC diagrams?

$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$



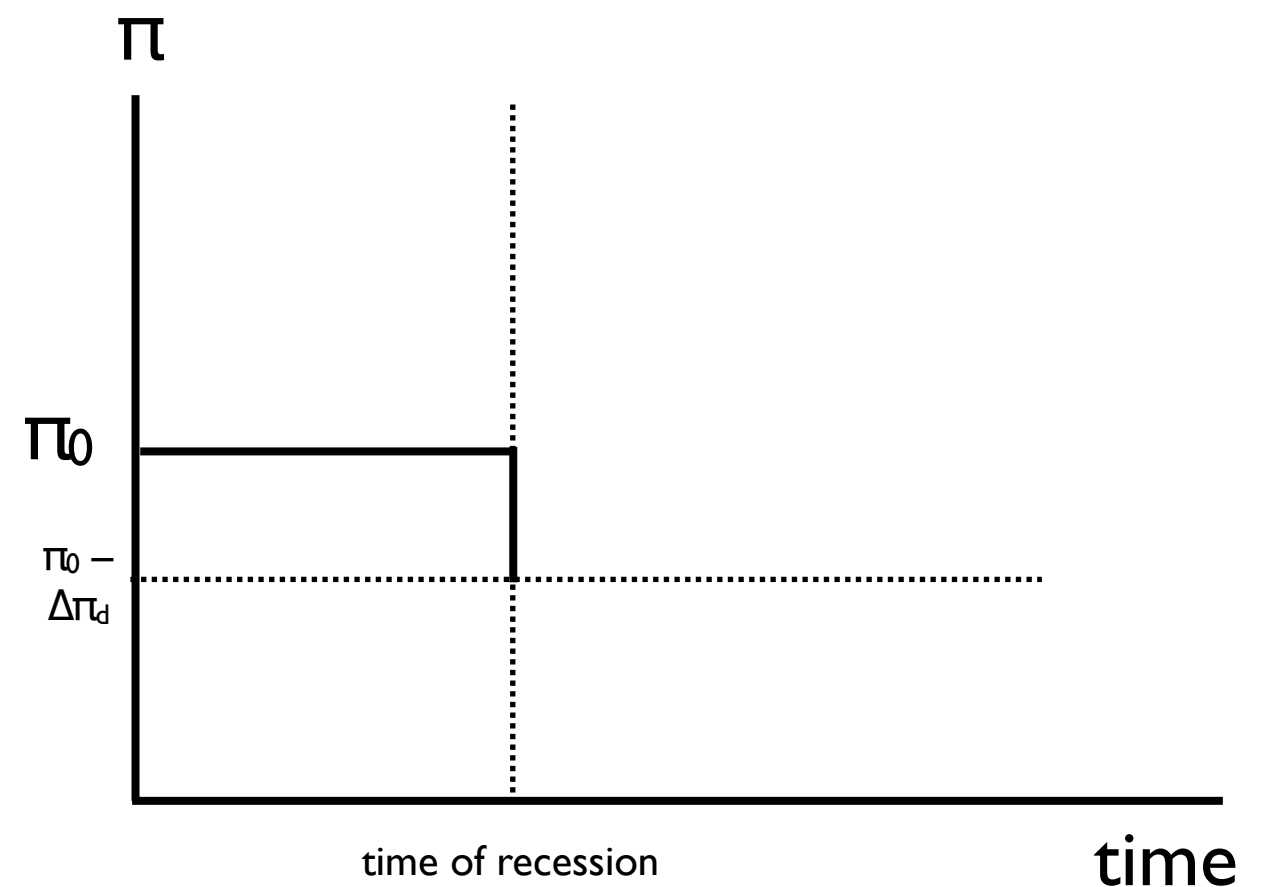
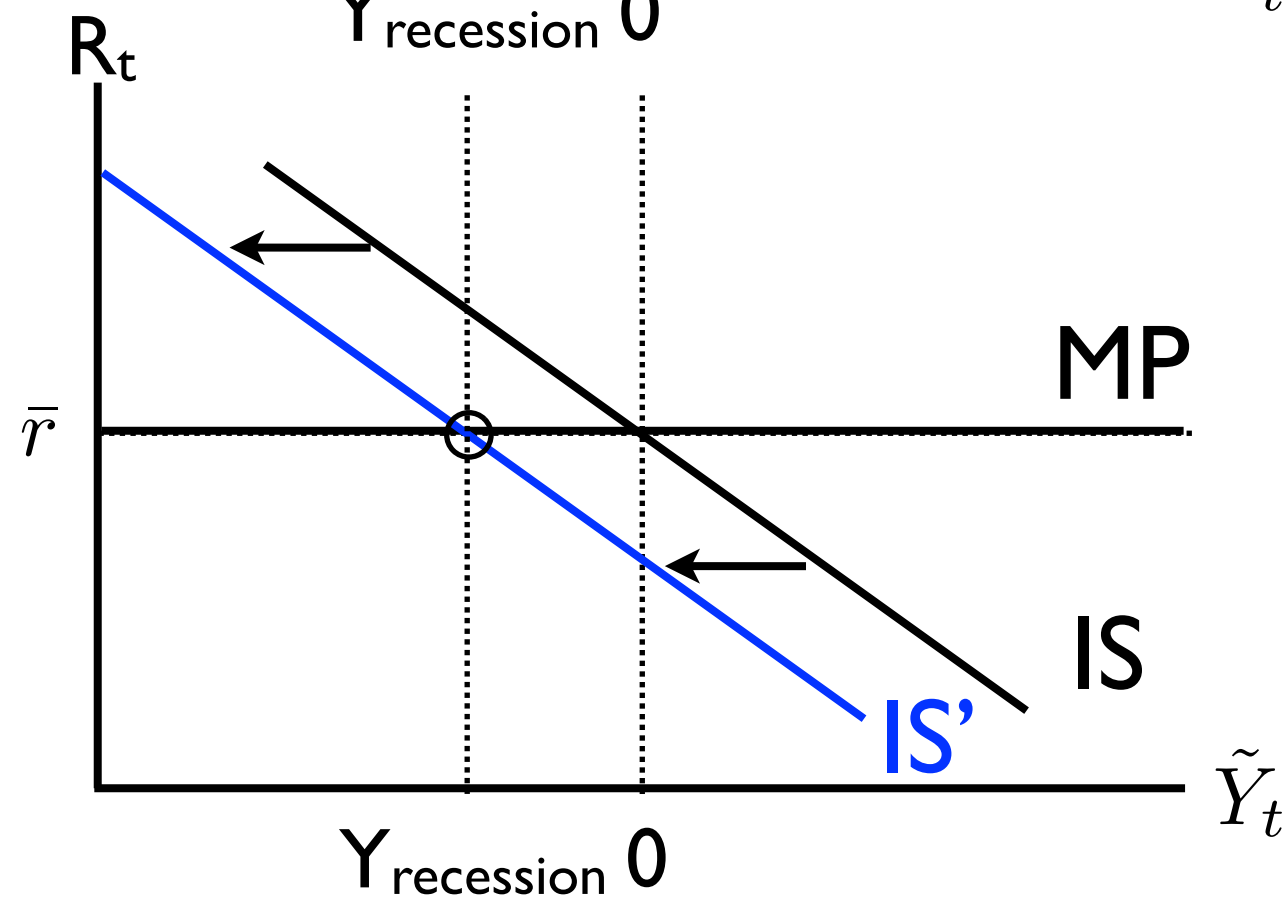
- This is a story about consumption spending that temporarily falls below its usual share of permanent income $\bar{a}_C = 0.64$
- When $\bar{a}_C$ falls, it pulls $\bar{a}$, the intercept of the IS curve, down with it.
- This is a *negative* aggregate demand shock of an amount equal to the reduction in $\bar{a}_C$
- It's a **leftward shift** in the IS Curve to IS'

Yet more practice with IS-MP & PC (2/3)

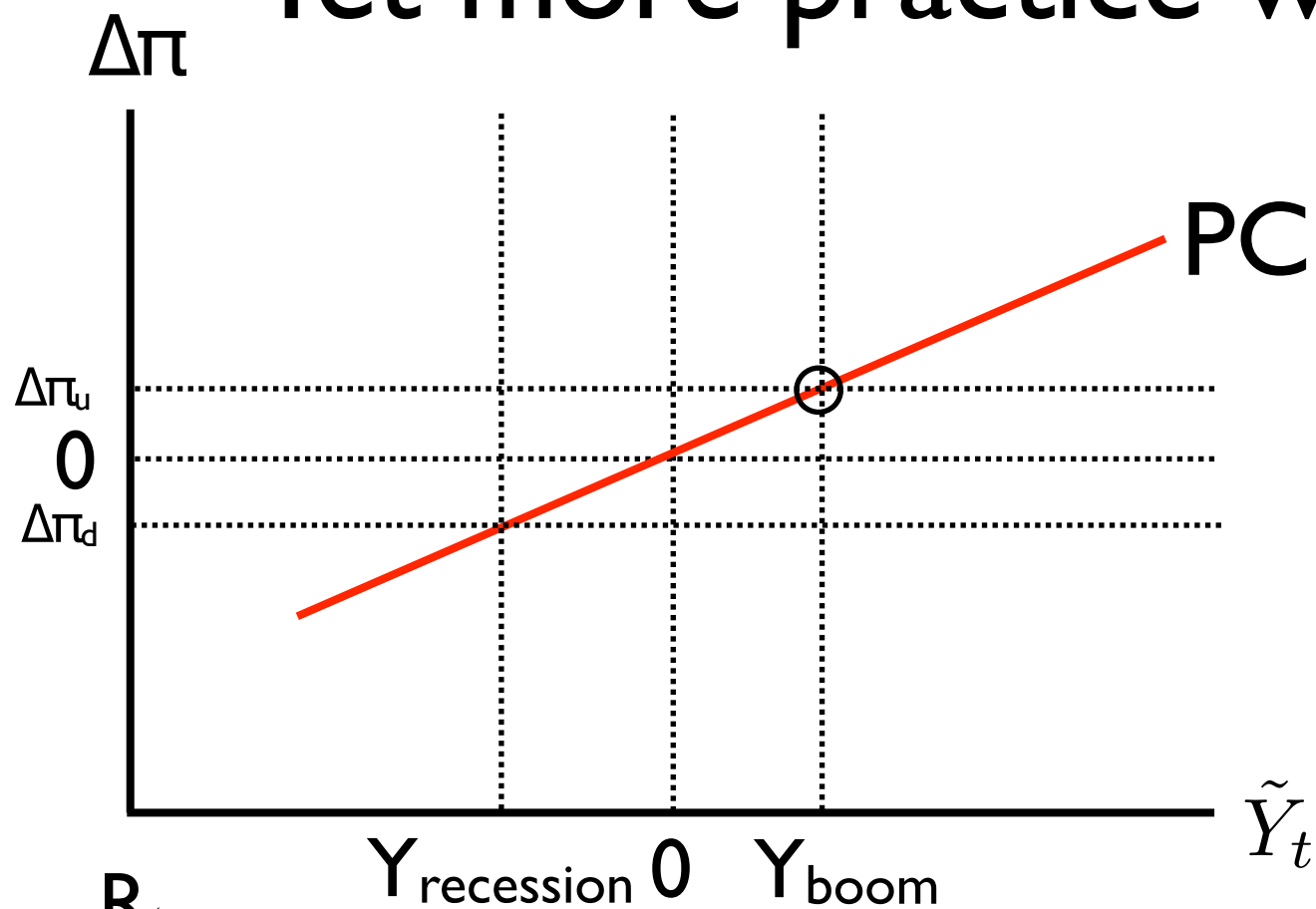


Q: Suppose consumers feel their houses, a major part of their wealth, have lost a lot of value, so they temporarily cut back on spending. What happens in the IS-MP or PC diagrams?

- The IS Curve has shifted to the left
- This produces a recession!
- This means that unless the Fed lowers the real interest rate R , inflation will decrease!

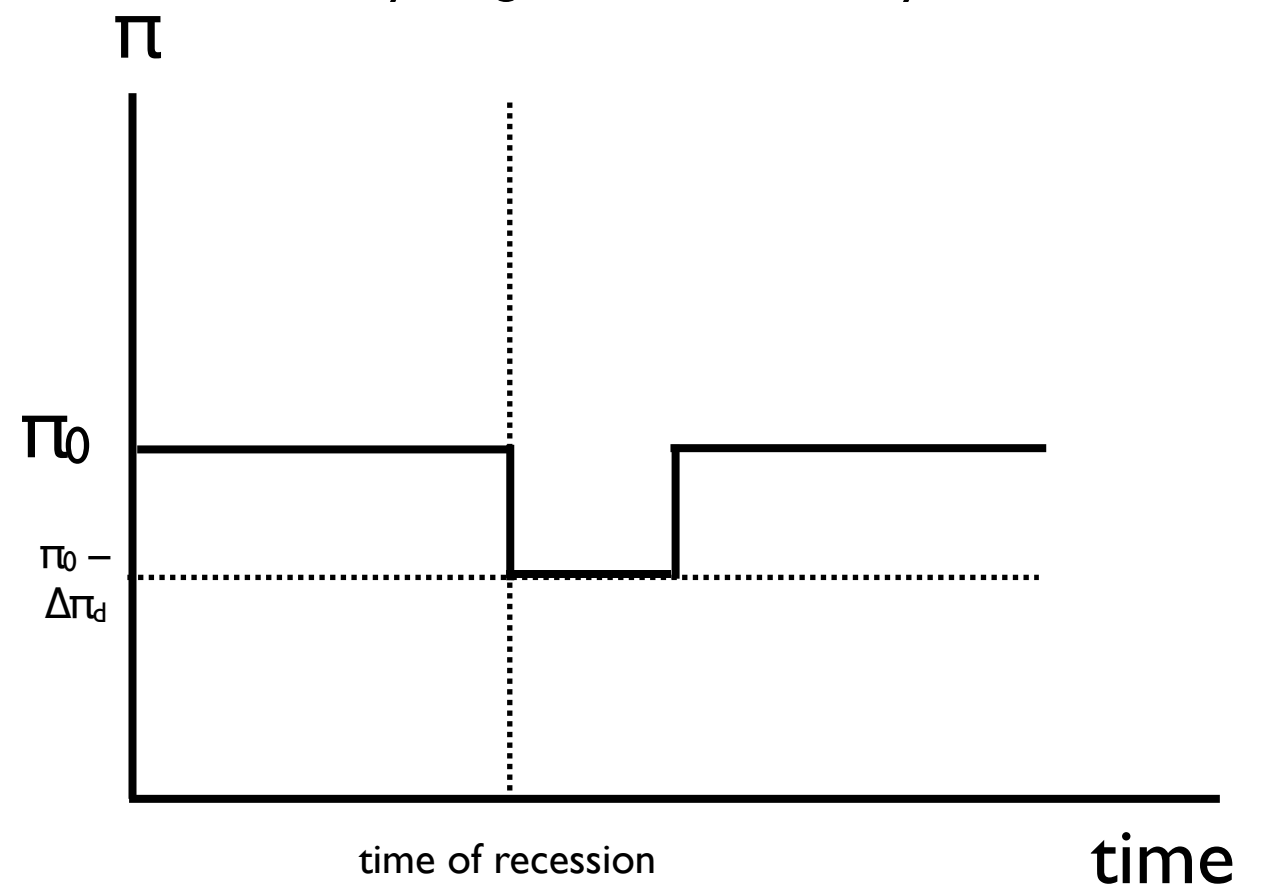
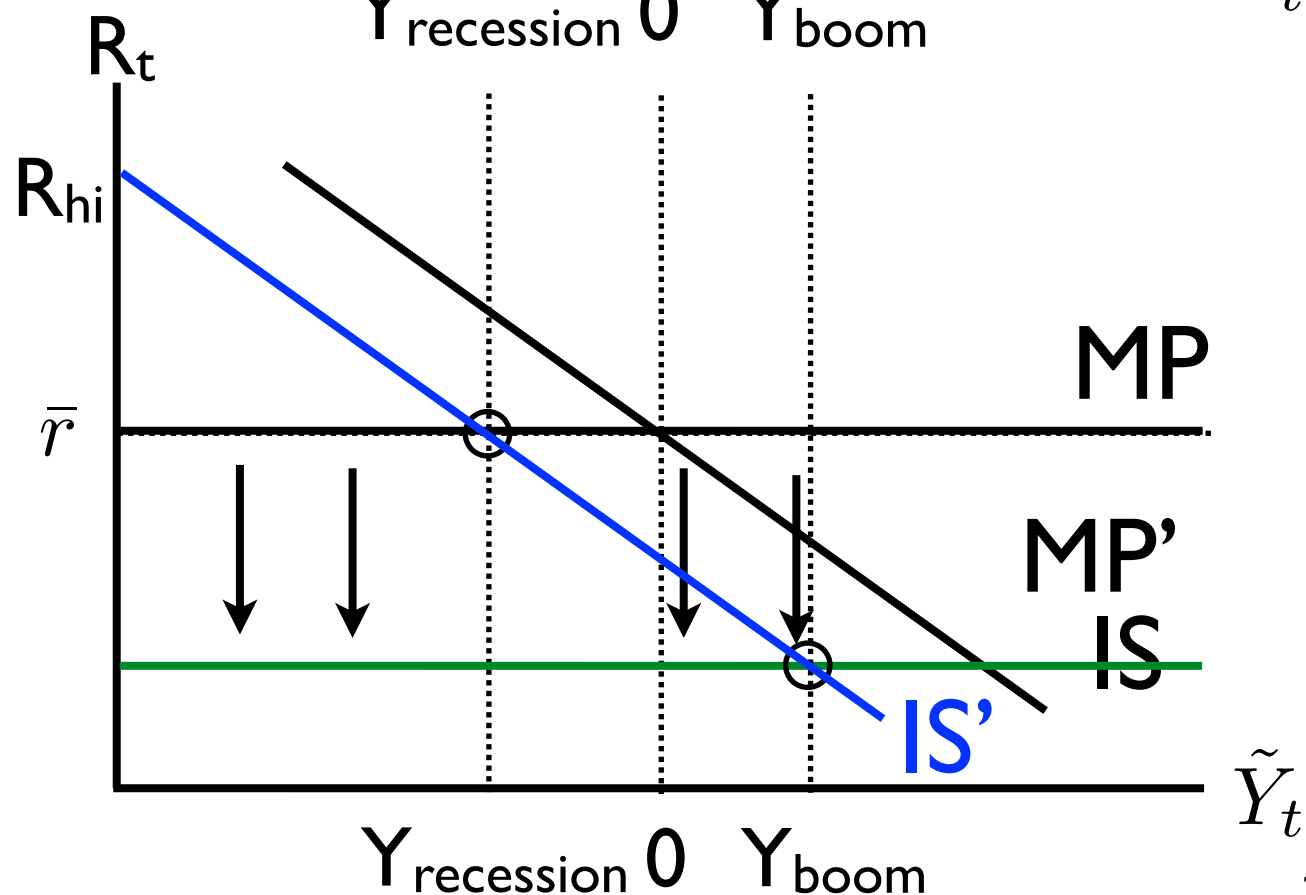


Yet more practice with IS-MP & PC (3/3)



Q: Suppose consumers feel their houses, a major part of their wealth, have lost a lot of value, so they temporarily cut back on spending. What happens in the IS-MP or PC diagrams?

- What should the Fed do if it wants to pull inflation back up toward π_0 ?
- Lower the interest rate R and cause a boom
- Inflation will rise, and if the Fed keeps R low for sufficiently long, π rises all the way back to π_0



How to think through the short-run model

- We care about
 - **Unemployment** and recessions: $\tilde{Y} < 0$
 - Inflation that is too high: $\pi > 0.02$ with target $\bar{\pi} = 0.02$
- Demand or supply considerations can affect either or both of these
- Exam or essay questions in Econ 100B are going to start with something about changes in demand or supply
- Then your task is to choose the right graph — either the IS-MP diagram or the Phillips curve PC diagram — and start modeling
 - What happens first
 - How the Fed reacts with monetary policy to try and obtain and maintain $\pi = \bar{\pi} = 0.02$

Why set monetary policy rules



Odysseus, bound to the mast of his ship, listens to the deadly song of the Sirens. British Museum, London [1843, 1103.31](#) 480-470 BCE

Homer and game theory

- Ben Bernanke, 2003, NABE conference, [“A Perspective on Inflation Targeting”](#)
- “I agree that human judgment should always be the ultimate source of policy decisions and that no central bank should — or is even able to — commit irrevocably in advance to actions that may turn out to be highly undesirable.
- “However, the intuition that more flexibility is always better than less flexibility is quite fallacious, a point understood long ago by Homer, who told of how Ulysses had himself tied to the mast so as not to fall victim to the songs of the Sirens.
- “More recently, the notion that more flexibility is always preferable has been pretty well gutted by modern game theory (not to mention modern monetary economics), which has shown in many contexts that the **ability to commit in advance often yields better outcomes.**”



Monetary policy rules

- A simple one in the textbook:

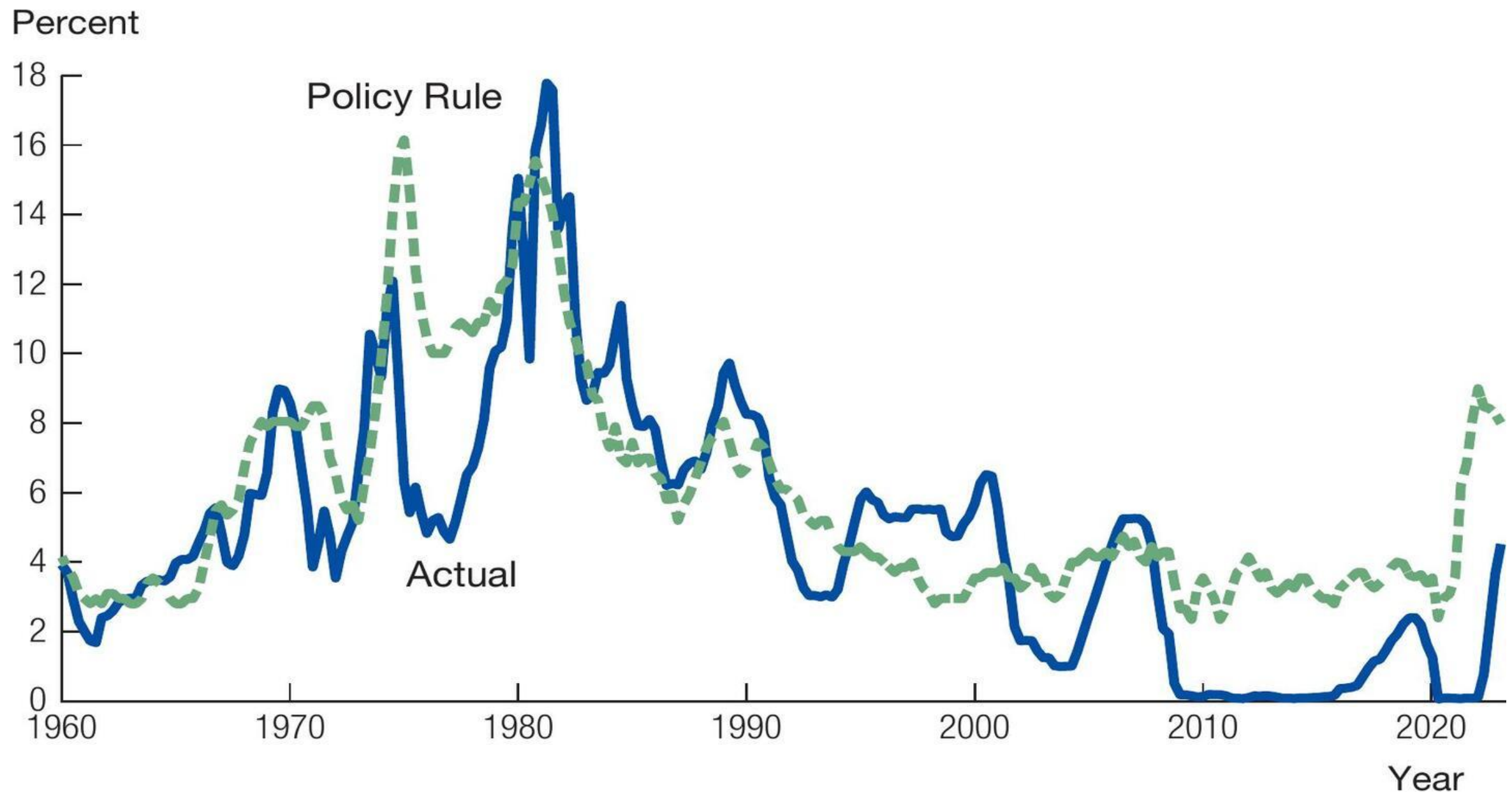
$$R - \bar{r} = \bar{m} (\pi - \bar{\pi}) \quad \text{with } \bar{m} = \frac{1}{2} \text{ and } \bar{\pi} = 0.02$$

- In words:
 - Raise R above the marginal product of capital, $\bar{r}$, when inflation is above 2%
 - Lower R below the marginal product of capital, $\bar{r}$, when inflation is below 2%
- Inspired by the **Taylor rule** suggested by Stanford economist [John B. Taylor](#) in the 1990s. His also included a $\tilde{Y}$ term

Why use a Taylor rule? Why not?

- It outlines precisely how the central bank will respond to inflation (and recessions)
- It ties the hands of the central bank, which promises that it will not respond otherwise
- It helps anchor expectations about the central bank's actions, and about inflation
- **An inflation target**
 - Can accomplish similar things
 - Avoids a problem in modern monetary policy: the **zero lower bound**

The textbook's graphic does not show the zero lower bound ever being breached



Source: The FRED database.

Shows a Taylor rule with $\bar{m} = 0.5$, $\bar{r} = 0.02$, and $\bar{\pi} = 0.02$

But [Nakamura, Riblier, and Steinsson \(2025\)](#) show the zero lower bound problem

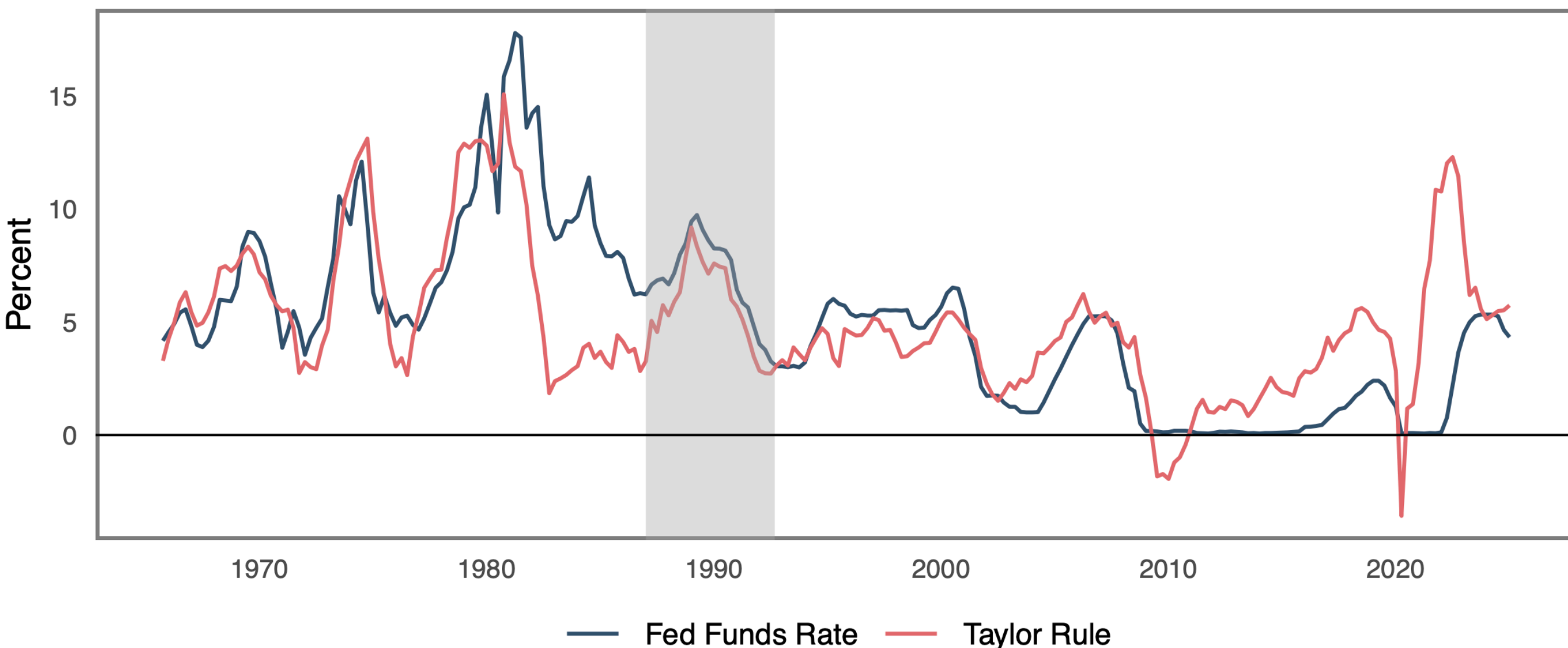


Figure 3: Original Taylor Rule with Real-Time Data

Note: The (light) red line shows the policy rate implied by the original Taylor rule—equation (1). We use real-time data for the GDP deflator and the Greenbook’s measure of the output gap (CBO output gap for last few years). The sample period is 1965Q4-2025Q1. The shaded area shows the sample period considered in [Taylor \(1993\)](#).

Shows a Taylor rule with $\bar{m} = 0.5$, $\bar{r} = 0.02$, and $\bar{\pi} = 0.02$

Case study of what not to do: IS-MP-PC and Turkey in 2022

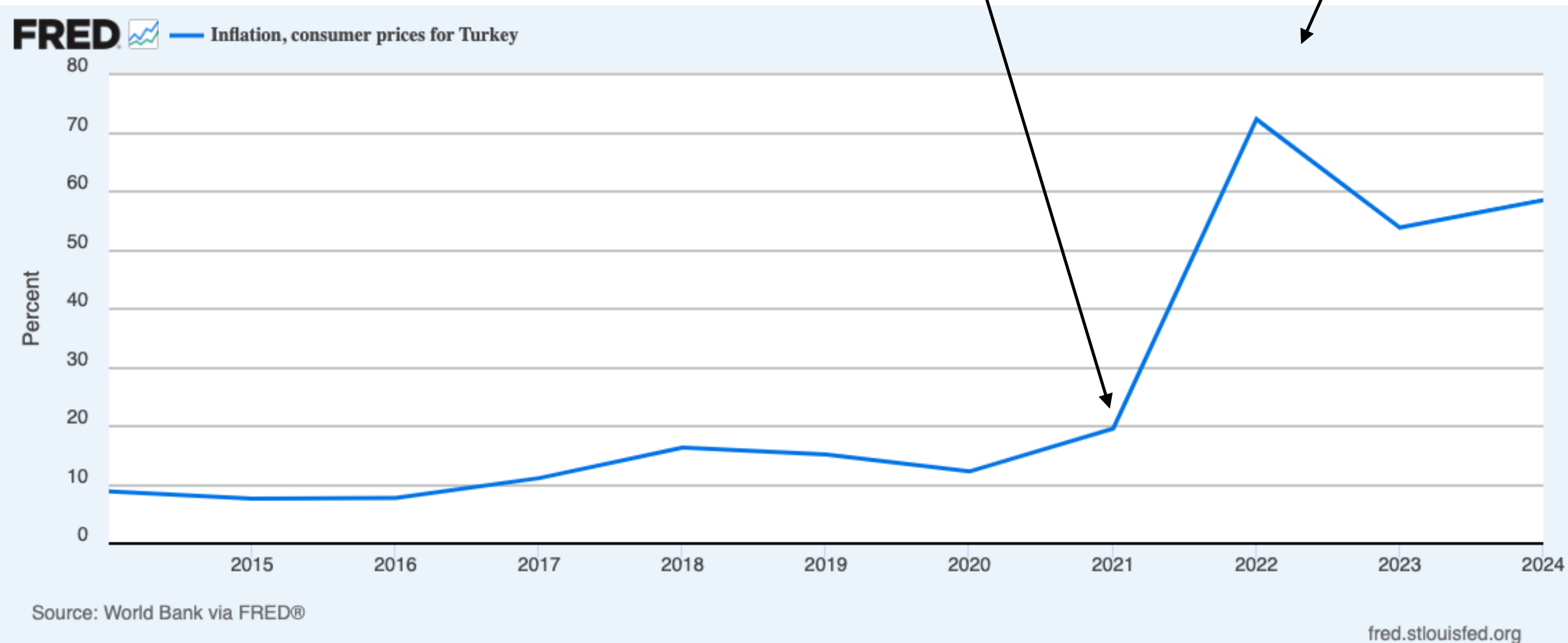
- History is littered with examples of bad monetary policies that follow takeovers of central banks by government or parties
- Notably, the inflation disaster in Turkey in 2022. See [WSJ 7/16/25](#)
- [Turkish President Erdogan fired four central bank chairs in less than two years](#)
- On March 19, 2021, Erdogan installed Sahap Kavcioglu; per WSJ,
 - A former member of parliament from Mr. Erdogan's Justice and Development Party
 - Columnist for the pro-government newspaper Yeni Safak
 - "Our country always lost out from high interest rates," he wrote in an article on Feb. 9

[WSJ reported:](#)

“Erdogan claimed that lowering interest rates would reduce inflation by cutting corporate borrowing costs. The idea contradicted widely accepted economic principles that low interest rates push prices up by spurring economic activity.”

March 19, 2021: President Erdogan fires the central bank chief for raising interest rates

2022: 72% annual inflation



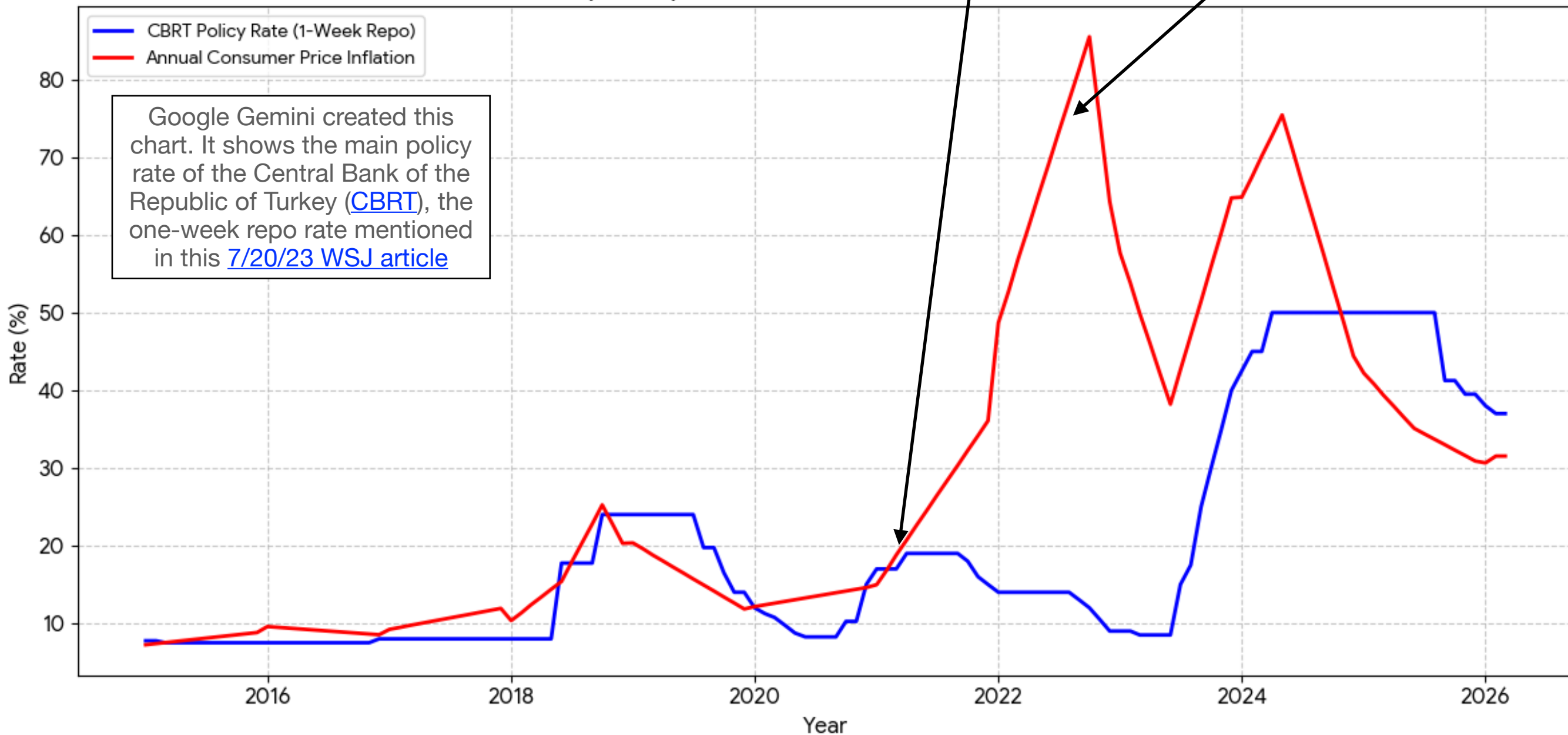
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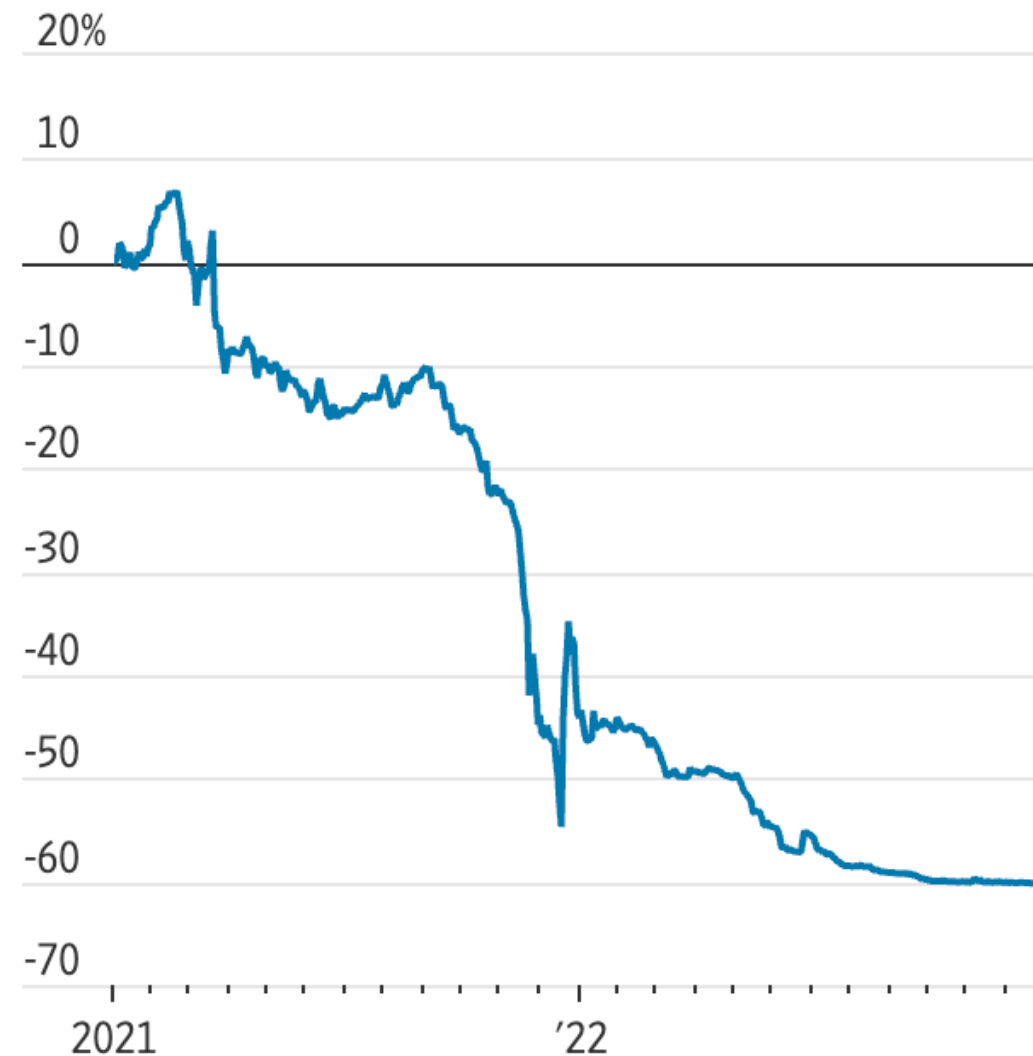
Turkey: Policy Rate vs. Annual Inflation (2015 - 2026)



Looking ahead: the exchange rate

- A big depreciation of the Turkish Lira followed
- Source: [WSJ 7/16/25](#)
- What drives exchange rates?
- Interest rate policies

Turkish Lira/U.S. Dollar



Source: Tullett Prebon